Lexitarianism: A Needs-Based Approach to Public Policy*

Alyssa Rusonik[†] ⓡ Larry Samuelson[‡] ⓡ Itzhak Gilboa[§]

August 18, 2026

Abstract

Utilitarian models typically do not attempt to capture distinctions between various determinants of well-being: they do not distinguish between wants and needs, and they assume that all determinants of well-being are commensurate. In contrast, welfare policies seem to reflect these distinctions. We propose to bridge the gap between utilitarian theory and economic practice by introducing *lexitarianism*: the maximization of a vector of lexicographically-ordered welfare functions, as applied to needs. We propose a formulation of the social choice problem based on the concept of a position and use it to axiomatize lexitarianism as well as its extensions to multiple time periods and uncertainty.

1 Introduction

1.1 The Importance of Utilitariainism

Economic theory typically asserts that interpersonal comparisons of utility are meaningless. Under the revealed-preference paradigm, "utility" is no more than a mathematical tool used to describe individual decision making, providing no basis for comparing the gains of one individual with the losses of another. Welfare comparisons are limited to cases of Pareto domination.

*We thank Maya Eden, Loren Fryxell, Margaret Meyer, Jacob Nebel, Marcus Pivato, and Peter Wakker, as well as participants of D-TEA, RUD, and the Normative Economics and Economic Policy seminar, for comments and references. Gilboa gratefully acknowledges Investissements d'Avenir ANR -11- IDEX-0003 / Labex ECODEC No. ANR - 11-LABX-0047, the AXA Chair for Decision Sciences at HEC. Rusonik gratefully acknowledges a state grant managed by the Agence Nationale de la Recherche under the Investissements d'Avenir programme (ANR-18-EURE-0005 / EUR DATA EFM).

†HEC, Paris. alyssa.rusonik@hec.edu

‡Yale University. larry.samuelson@yale.edu

§HEC, Paris. gilboa@hec.fr

In contrast, public policies can rarely be selected on the basis of Pareto domination. Decisions have to be made that involve winners and losers. Fortunately, people seem to share a moral intuition about many such decisions, and this intuition is reflected in public policy. For example, most people tend to agree that it is easier for the rich than for the poor to part with a given amount of money, and so taxation is typically progressive.

The most widely-used model for such intuition, and the attendant decisions, is utilitarianism, dating back to Bentham (1780/1789) and offering an attractive combination of tractability, completeness, and formal elegance. For example, the United States Office of Management and Budget (2023) prescribes that cost-benefit analysis be routinely used in the evaluation of government projects, with Section 10 devoted to procedures for weighting and aggregating individual effects. Utilitarianism has accordingly received extensive attention from economic theorists, beginning with Harsanyi's (1953, 1955) celebrated axiomatic derivations. Interest in the underpinnings of utilitarianism has not waned, as evidenced by recent contributions such as Bossert and Kamaga (2020), Li and Wakker (2024), Echenique and Valenzuela-Stookey (2024), and Karni and Weymark (2024).

1.2 Revealed Utilitarianism

The theoretical conception of utilitarianism is straightforward. One identifies the bundle x^i of goods allocated to each person i, identifies the utility $v^i(x^i)$ that person i attaches to the bundle, and adds the (possibly weighted) utilities. In contrast, the version of utilitarianism implicitly embedded in social policy formation typically exhibits three features not captured by this conception.

First, public policy applications of utlitarianism take a sympathetic rather than empathetic approach.[1] The standard empathetic approach in welfare economics asks the planner to maximize the sum of the individuals' utility functions. Each utility function v^i appearing in the theoretical conception of utilitarianism is a bespoke description of individual i's preferences. This raises

[1]See Fontaine (1997, 2001) and Bloom (2016) for the distinction between empathy and sympathy.

a host of practical difficulties and opportunities for manipulation. The utilitarian allocation of resources will depend on which of the unlimited collection of behaviorally equivalent utility functions one chooses to represent each individual's preferences. More importantly, as we illustrate in Section 5.1, conceptual difficulties arise even before one turns to the mechanics of implementation. By contrast, public policy discussions typically adopt a sympathetic approach that views people as having the same objectively-measurable needs, addressed to the same extent by various consumption bundles. In effect, a common utility function u is used to evaluate the well being of every person. We argue in Section 5.1 that the sympathetic utility function u reflects the same sort of social consensus that undergirds public policies such as progressive taxation.

Second, utilitarianism implicitly assumes that all determinants of utility are commensurate. If a utilitarian social welfare function recognizes some people's enjoyment of opera, sufficiently many opera performances would compensate for a small share of starving children—in a way that many would find counterintuitive as a guide for public policy. Public policy discussions typically avoid such comparisons, while they are built into the structure of utilitarianism.

Third, the practical utilitarian calculus draws a distinction between needs and wants, and focuses on needs. Distinctions between needs and wants go back at least to Epicurus (see Cooper, 1998). In economics, Marshall (1890, Book II, Chapter III) wrote "It is common to distinguish necessaries, comforts, and luxuries". Admittedly, it is not always easy to draw these distinctions. Does one want a cold beer on a hot summer day only because it quenches thirst? Is the pleasure it yields then simply than the cessation of pain, i.e., the satisfaction of a need? Schopenhauer (1818) suggested that pleasure is no more than the absence of pain, potentially allowing an apologist to rationalize any want as a need. For example, is the desire to listen to a concert an expression of a want, or a spiritual or cerebral need?

While the precise boundary is difficult to draw, there are plentiful examples in which most people would agree on distinctions between needs and wants. If a person were to say that she needs to have a space trip very much like another person needs to have a meal, most of us would reject the analogy as ridiculous or outrageous. Psychology draws an analogous distinction between pain avoidance and pleasure seeking in a rather clear way (see Elliot and Covington, 2001). Moreover, the distinction between needs and wants is reflected in public and economic policy: many countries provide some level of nutrition, health care, and shelter to the needy, yet no country has a budget for space trips of its citizens as a form of entertainment. On the contrary, many countries even levy taxes on luxury goods. Public policy thus appears to recognize the intuitive distinction between needs such as nutrition and health care, and wants such as space tourism and luxury goods.

The focus on needs and the sympathetic approach are complementary. "Wants" are inherently subjective—different people want different things—and hence require an empathetic language to be adequately described. A sympathetic approach to an aggregation of wants fails to get off the ground, stymied by being tied to a system of measurement too coarse to capture the diversity of wants. In contrast, one might reasonably invoke arguments analogous to those of Stigler and Becker (1977) to argue that people have relatively common needs. These common needs can be captured by a sympathetic utility function, rendering an empathetic approach redundant.

We thus see a gap between the the canonical model of utilitarianism and the utilitarianism implicitly embedded in policy formation. The former takes an empathetic approach to commensurate wants. The latter takes a sympathetic approach to potentially incommesnsurate needs.

1.3 The Literature

The literature contains a number of responses to this gap. At the opposite extreme of making everything commensurate, Rawls (1971) suggests that we focus on primary goods and consider only the individual with the least desirable allocation of primary goods. This extreme rigidity can be alleviated slightly by the leximin refinement (Hammond, 1976), which suggests that, given indifference of the worst-off individuals, we should still care about the ones next up on the utility scale. However, there is still no trade-off between the least advantaged and any other members of society. For example, in a society where the poorest individual lives on 400 grams of rice a day, and the rest on 450 grams, a leximin social planner would reject a proposal to reduce the poorest individual's ration from 400 to 380 grams a day, even if doing so promised abundance for everyone else. The maximin and leximin criteria are thus often viewed as uncomfortably extreme in their refusal to make certain tradeoffs.

Somewhat more flexibly, sufficientarianism (Frankfurt, 1987; Alcantud, Mariotti, and Veneziani, 2022; Bossert, Cato, and Kamaga, 2023) distinguishes between people who do and do not attain a level of well being that might be interpreted as satisfying basic needs. Once individuals "have enough", further gains do not matter. The social planner maximizes the number of individuals who have enough, without drawing distinctions among the levels of well-being. It seems counterintuitive not to prefer that individuals who have enough would have more, if they can get more without pushing anyone else below the threshold. It seems even more counterintuitive that the social planner would be indifferent to improving the lot of individuals who do not have enough.

While maximin/leximin and sufficientarism are starkly different, they have a common feature of incommensurability: the former focus on the worst-off individual as incommensurately more important than all those who are better off, while the latter views the attainment of a utility threshold as incommensurately weighter than any differences in utility on either side of this threshold. More generally, ideas of incommensurability have been discussed in the philosophical literature, in varying contexts. Nebel (2022) writes, "The idea that some goods might be lexically superior to others is not new, of course. It is at least as old as Mill (1863); some attribute it to Aristotle." Specifically, the notion of lexicographic orderings has been used to cope with Parfit's (1984) Repugnant Conclusion (Thomas, 2018; Nebel, 2022; Thornley, 2022). More formally, Narens (1974), Carlson (2010), and Pivato (2014) axiomatize lexicographic decision rules. The latter provides a representation of a preference relation by a non-Archimedean abelian group, which can also be used to derive a representation that we refer to as lexitarian. Moreover, some of his suggested applications are for social choice, and are in line with our motivation. (See also Pivato and Tchouante, 2024). However, our model differs from Pivato's in focusing on the notion of "positions"which capture the level of satisfaction of needs. The axiomatic derivation we offer is not only mathematically different from Pivato's, but (arguably) simpler and more transparent. In particular, it can more readily serve as a theoretical basis for the elicitation of moral preferences.

By counting occupants of positions, our lexitarian criterion allows for comparisons between societies of the same or different sizes. Whether the population is fixed or variable is an important consideration in the population ethics literature, which draws a distinction between person-affecting views and impersonal approaches.[2] The former (see, e.g., Narveson, 1973, for a foundational work) maintains that an outcome can only be better or worse if it is better or worse for individuals whose existence is not conditional on the various policy alternatives under consideration, while the impersonal approach includes choice-contingent variable populations. The person-affecting approach is immune to one of the primary critiques of the impersonal approach: the Repugnant Conclusion and its variants (Parfit, 1984), which show that, if alternative-contingent populations are incorporated, a sufficiently-large, barely-subsistent society can always be constructed to dominate a smaller and more prosperous society. On the other hand, person-affecting views have been criticized as giving rise to the "Non-Identity Problem" and other difficulties in evaluating the moral significance of creating new lives (Parfit, 1984), which arise from ignoring the welfare of the conditionally-existent.[3] Our accommodation of variable population sizes places our analysis closer to the impersonal view of population ethics. The lexicographic criterion can be used to mitigate some of the potentially-problematic conclusions of mere additions (indeed, as mentioned above, such ideas have been proposed by Thomas, 2018; Nebel, 2022; Thornley, 2022).

[2]See, e.g., Carlson (1998), Arrhenius (2000), Roberts (2011), Boonin (2014), and Frick (2020). The issue has also been discussed in social choice theory by Blackorby and Donaldson (1984), among others.

[3]Person-affecting approaches are also more vulnerable to intransitivities.

1.4 Lexitarianism

We suggest that the social welfare reasoning implicitly embedded in common public policy discussion and implementation is most effectively captured by a "lexitarian" criterion: the lexicographic maximization of the sums of several (ordered) sympathetic utility functions, each of which captures the degree to which different commensurate needs are satisfied. As will be clear from the discussion, we believe that the lexitarian model captures people's moral intuition better than the utilitarian one. This is a positive statement about people's ethical preferences. If accepted, we also propose the lexitarian model as a framework for normative discussions.

As a point of comparison, under the standard utilitarian approach to social welfare, each of the members $1, \ldots, K$ of a population $\mathcal{K}$ is characterized by an allocation drawn from a set X of consumption bundles. Aggregate well-being is given by a function $V : X^K \to \mathbb{R}$ defined by

$$V(x^1, \ldots, x^K) = \sum_{i \in \mathcal{K}} v^i(x^i),$$

where v^i is person i's utility function and x^i completely specifies (only) person i's consumption.

To bring the features of practical utilitarianism described in Section 1.2 into the model, we adopt a sympathetic framework in which the social planner uses a single utility function $w : X \to \mathbb{R}$ to evaluate the outcome for every member of the population. If we were to implement this departure alone, aggregate welfare would be given by a function $W : X^K \to \mathbb{R}$ of the form

$$W\left(x^1, ..., x^K\right) = \sum_{i \in \mathcal{K}} w\left(x^i\right). \tag{1}$$

Next, we assume that the social planner has a collection of functions $w_j : X \to \mathbb{R}$, and consider the lexicographic ordering induced by successively calculating (1) for the functions $(w_1, ..., w_m)$. The interpretation is that w_j captures the planner's evaluation of the consumption of goods at level j. The planner considers tradeoffs among goods in level j, but is unwilling to trade them off against the goods appearing at other levels. For example, we can imagine the function w_1 as evaluating the consumption of goods ensuring health and nutrition, while w_2 evaluates the consumption of goods leading to self-fulfillment and personal expression. Such a model would allow one to say that providing medication to sick people may justify lowering the level of nutrition of some others, while also indicating that art classes for some individuals should not be traded for the nutrition of others. Further, one may use Maslow's (1943) hierarchy as a guide for defining utility functions. Indeed, lexitarianism as proposed here can be thought of as "Maslow meets Bentham": the Benthamite idea of additive aggregation is preserved within each level, but the hierarchy of levels follows an analog of Maslow's pyramid of needs.

Finally, we propose a model in which the primitive is a set $\mathcal{P}$ of "positions" people may occupy. In our intended interpretation, a position $p \in \mathcal{P}$ specifies the degree to which a certain need is satisfied. For example, position p might identify a person as being free from hunger. The objects of choice are population profiles, which describe how many individuals are at each position $p \in \mathcal{P}$. Formally, a profile is a "counter vector" I, where $I(p) \geq 0$ indicates how many individuals are at position p. The overall number of individuals in a profile (ranging over all positions) is finite, but we impose no a priori bounds—neither on the number of positions that may be occupied, nor on the number of individuals at each position. A given individual may appear in a profile I as part of the count $I(p)$ at many positions p, corresponding to different needs relevant for the individual. In the position-based approach to standard utilitarianism, the planner is characterized by a utility function $u : \mathcal{P} \to \mathbb{R}$ and maximizes

$$I \cdot u = \sum_{p \in \mathcal{P}} I(p)\, u(p) . \tag{2}$$

With enough freedom in the definition of a position, the position-based and person-based approaches are equivalent. To see this, consider first the sympathetic utilitarian model with a set of bundles X and a utility function $w : X \to \mathbb{R}$ so that $\bar{x} = (x^1, \ldots, x^K) \in X^K$ is evaluated by $W(\bar{x}) = \sum_{i \in \mathcal{K}} w(x^i)$. We can define $\mathcal{P} = X$ and $u = w$, and, given $\bar{x} = (x^1, \ldots, x^K) \in X^K$, define a profile $I_{\bar{x}} : \mathcal{P} \to \mathbb{Z}_+$ by $I_{\bar{x}}(p) = |\{\, i \mid x^i = p \,\}|$. We will then have

$W(\bar{x}) = I_{\bar{x}} \cdot u$. Conversely, suppose we are given a set of positions $\mathcal{P}$ and a function $u : \mathcal{P} \to \mathbb{R}$, so that a profile $I : P \to \mathbb{Z}_+$ is evaluated by $I \cdot u$. We can set $X = \mathcal{P}$ and $w = u$, and, given a profile I, define $K_I = \sum_{p \in \mathcal{P}} I(p)$ and $\bar{x}_I = (x^1, \ldots, x^{K_I}) \in X^{K_I}$ with $I(p) = |\{ i \mid x^i = p \}|$. It will follow that $I \cdot u = W(\bar{x})$.[4] Further, a similar equivalence holds when the social planner is allowed to have a lexicographic utility function.

A sufficiently rich set of positions can embed all determinants of utility, and even allow for utilitarian aggregations that violate anonymity.[5] This, however, is not our intended interpretation. Rather, we think of a position as summarizing a certain aspect of well-being, such as level of nutrition or state of health. It is part of the sympathetic social planner's evaluation of a social alternative, and has a built-in anonymity assumption. We believe that the position-based model offers a useful perspective on social welfare and on underlying axioms. Specifically, the lexitarian model will be simply axiomatized in this framework, with the lexicographic order over utility functions naturally derived from the social planner's preferences.

1.5 Preview

Section 2 presents the basic model and a few general axiomatizations. In Section 3 we explore several ways in which the formal model can be used to capture the notion of needs, and explain our preferred modeling choice. Section 4 discusses the specification of the model to incorporate uncertainty and different time periods. Section 5 concludes with a discussion providing some background for our modeling choices and exploring the degree to which the model captures public policy.

[4] An empathetic utilitarian model, in which each individual i has a potentially distinct utility v^i, can also be embedded in the positions model, if the set of positions is defined to be the values of the functions v^i.

[5] Indeed, if a position is defined as "person i experiences x", it can only be occupied by a single person.

2 Model and Results

Let there be a set of *positions* $\mathcal{P}$. The set of *profiles* is

$$\mathcal{I} = \left\{ I : \mathcal{P} \to \mathbb{Z}_+ \;\middle|\; \sum_{p \in \mathcal{P}} I(p) < \infty \right\}.$$

For $I \in \mathcal{I}$, the *support* of I is $supp(I) \equiv \{ p \in \mathcal{P} \mid I(p) > 0 \}$. Thus, the set of positions $\mathcal{P}$ may be infinite, but each profile I has a finite support. Algebraic operations are performed on $\mathcal{I}$ pointwise so that $nI + mJ$ is well-defined for $n, m \geq 0$ and $I, J \in \mathcal{I}$. $0 \in \mathcal{I}$ has its natural meaning. Similarly, for $I, J \in \mathcal{I}$ the inequality $I \geq J$ is read pointwise. For $p \in \mathcal{P}$, $\mathbf{1}_p \in \mathcal{I}$ denotes the indicator function for p. For a subset of positions $A \subset \mathcal{P}$, we denote by $I_A \in \mathcal{I}$ the corresponding profile that vanishes outside A, i.e. $I_A = \sum_{p \in A} I(p) \mathbf{1}_p$.

For a function $u : \mathcal{P} \to \mathbb{R}$, denote

$$I \cdot u = \sum_{p \in \mathcal{P}} I(p) u(p)$$

which is well-defined as I has a finite support. For $u : \mathcal{P} \to \mathbb{R}$ and $A \subset \mathcal{P}$, $u_A : A \to \mathbb{R}$ is the restriction of u to A.

Let there be given a binary relation $\succsim \subset \mathcal{I} \times \mathcal{I}$. The symmetric and asymmetric parts of $\succsim$ will be denoted, as usual, by $\sim$ and $\succ$, respectively.

We impose four axioms. First,

A1 Weak Order: $\succsim$ is complete and transitive.

The completeness axiom is notoriously problematic, especially in the context of social choice. Its main justification is, as always, that decisions need to be made, and that it is better to bring them forth and see them in the model rather than to leave them unspecified. [6]

Transitivity is one of the most basic axioms of choice models, and we do not have anything to say about it that is specific to the current model. The main reason to highlight it is that some of the literature on the "Repugnant

[6]There are papers which opt for quasi-orderings instead, including, for instance Parfit (1982), which introduces indeterminacy via ranges of utility specified for added individuals.

Conclusion" suggests dropping this axiom (see Temkin, 1987, Persson, 2004, and Rachels, 2004). We tend to believe that giving up on transitivity is a high price to pay, both theoretically and practically, and that it is worthwhile to explore the possibilities that respect this axiom.

Our next axiom is the main driving force behind the additive representation:

A2 Union: For all $I, J, K \in \mathcal{I}$, $I \succsim J$ iff $I + K \succsim J + K$.

Consider first an interpretation of the model where a position summarizes all the information about an individual, so that each individual is counted in a profile at one position only. In this case, suppose that the social planner can choose between one policy, leading to a population profile I, and another, which will result in a population profile J. There is also another location, where the population profile will be K, irrespective of the current policy choice. The Union axiom states that the preference between I and J should not change if each is combined with the population whose profile is K. Clearly, it is very similar to axioms of separability across populations, and to Savage's P2 (Savage, 1954). It also bears resemblance to von Neumann and Morgenstern's (1944, 1947) Independence axiom, basically stating that a subpopulation that is common to two profiles can be ignored.

The Union axiom is, however, more questionable when a position is interpreted as describing only some aspects of an individual's well-being. For example, if we think of food and self-fulfillment as incommensurate needs, and if each individual appears in a food-position as well as in a self-fulfillment-position, the axiom states that we ignore inequality in food distribution when we consider inequality in self-fulfillment. We return to discuss these modeling choices in Section 5.

The third axiom is a monotonicity axiom, indicating that the central planner prefers that any population survives rather than dies:

A3 Monotonicity: For all $I \in \mathcal{I}$, $I \succsim 0$.

This axiom is substantive. Parfit (1984) criticizes aggregative welfare approaches for giving rise to the "Repugnant Conclusion" that a sufficiently-large society of people just barely clinging to existence would be preferred to a smaller society of people flourishing in abundance.[7] In its insistence that all lives are valuable, Axiom A3 raises similar possibilities.

One of the more influential attempts to avoid the repugnant conclusion is *critical level utilitarianism* (Blackorby, Bossert, and Donaldson, 1995; 1997), which posits the existence of a critical threshold below which an individual's well-being does not contribute to aggregate welfare.[8] The interpretation is that lives characterized by utilities below the critical level are not worth living. We could similarly incorporate critical levels into our formulation by dropping A3 and allowing the utility function $u(p)$ to assume negative values. In such a model, the critical level 0 would be endogenously determined as part of the description of social preferences.[9]

We choose a more conservative approach here, according to which the central planner is not permitted to decide that certain lives are not worth living. This conservatism is firmly rooted in the sympathetic paradigm: while an empathetic social planner has the freedom to consider an individual's subjective

[7]Asheim and Cato (2026) show that the Repugnant Conclusion, also known as the "Mere Addition Paradox" and the related "Very Sadistic Conclusion" are quite general, and do not depend on comparability assumptions such as anonymity or Archimedeanity. The paradoxical nature of the problem is contested: some philosophers have disputed the "repugnancy"of the conclusion (Huemer, 2008; Tännsjö, 2002; Gustaffson, 2022), while others have considered this as revelatory of a fundamental flaw to aggregation of welfare in ethics and utilitarianism (Temkin, 2012).

[8]Several directions have emerged in the population ethics literature to address the challenge of the repugnant conculsion, including variants of average utilitarianism (see Harsanyi (1955) and Sikora (1975), and see Parfit (1984) and Arrhenius (2000) for critical assessments), rank-dependent approaches (Zuber and Asheim, 2012); prioritarianism (Adler, 2012; Parfit, 1997); and various threshold-based modifications, including leximin principles (Hammond, 1976), utilitarianism dampened by population (Hurka, 1983), maximin welfare orderings (Bossert, 1990), and critical thresholds below which individuals' well-being does not contribute to aggregate welfare (Blackorby, Bossert, and Donaldson, 1995; 1997). Recent contributions that deal with the repugnant conclusion and related problems in a utilitarian framework include Pivato (2020) and Nebel (2024). See Broome (2004) for a discussion of the merits and drawbacks of aggregative welfare approaches in general, and non-unique critical thresholds in particular.

[9]If we deal with the special case of utilitarianism, namely, when the lexicographic ordering is trivially defined by a single function, one only needs to drop A3. For the counterpart of the lexicographic result, the definition of "commensurate", which follows, would have to be modified.

assessment of their own life, the sympathetic planner must commit to a single normative threshold for all individuals. We view the only tenable moral standard for prescriptive public policy—especially given that individuals may disagree with the planner—as that of Axiom 3. In other words, whether an individual's life is worth living is not within the legitimate purview of public policy, nor of the sympathetic paradigm more broadly.

For our final axiom, we need the notion of commensurability:

Definition 1 *For $p, q \in \mathcal{P}$, p and q are* commensurate *if there exists $k \geq 1$ such that $k\mathbf{1}_p \succsim \mathbf{1}_q$ and $l \geq 1$ such that $l\mathbf{1}_q \succsim \mathbf{1}_p$.*

Thus, two positions p and q are commensurate if a profile with sufficiently many people at position p is at least as desirable as a profile with a single person at q, and vice versa. Intuitively, the order $\succsim$ allows trade-offs over commensurate positions. Note that the intuition for this definition relies on Monotonicity: it is implicitly assumed that being at each position is a good thing, and thus, unless p and q are incommensurate, a large population at one position should eventually be a better society than a single person at the other. Conversely, should one believe that a position p describes "life not worth living", a larger population at position p would only make the resulting society worse in the eyes of the central planner.

If positions p and q are not commensurate, given completeness, it must be the case that one is infinitely more desirable than another. For example, being healthy may be considered incommensurably more important than having a sense of self-fulfillment. Importantly, the notion is endogenous: it is defined by the social planner's preference relation $\succsim$. Our model does not dictate which positions are incommensurably more important than others, nor that there be such pairs to begin with. With commensurability thus endogenously-defined, we proceed to state the last axiom.

A4 Restricted Archimedeanity: For all $I, J, K, L \in \mathcal{I}$, if $p, q \in supp\,(I) \cup supp\,(J) \cup supp\,(K) \cup supp\,(L)$ are commensurate and $K \succ L$, then there exists $n \geq 0$ such that $I + nK \succ J + nL$.

If we ignore the commensurability condition, the axiom is rather standard: it says that, if profile K is strictly preferred to profile L, then taking the union of any profiles I and J with sufficiently many copies of K and L, respectively, would result in a preference for the I-with-K's population over the J-with-L's one. The axiom states that the difference between J and I cannot be infinitely weightier than that between K and L. The fact that it is restricted to some quadruples (I, J, K, L) is at the heart of lexitarianism: the motivation for our model is precisely the fact that some differences are *not* commensurable. However, the axiom states that incommensurability isn't arbitrary, and it is basically defined by positions: if all the positions involved in (I, J, K, L) are pairwise commensurate—according to the planner's own preferences—then the axiom holds.

Next, we define the notion of numerical representation we seek:

Definition 2 *A* lexitarian representation *of $\succsim$ is a pair $\left(\succsim^b, u\right)$ where $\succsim^b$ is a weak order over $\mathcal{P}$ and $u : \mathcal{P} \to \mathbb{R}_+$ such that, for all $I, J \in \mathcal{I}$, $I \succ J$ iff there exists a $\succsim^b$-equivalence class $A \subset \mathcal{P}$ such that*

$$I_B \cdot u = J_B \cdot u$$

for every $\succsim^b$ equivalence class for which $B \succ^b A$ and

$$I_A \cdot u > J_A \cdot u,$$

where $\sim^b$ and $\succ^b$ are the symmetric and asymmetric parts of $\succsim^b$, respectively.

Note that, for every $I, J \in \mathcal{I}$, we have $I_B = J_B = 0$ for all but finitely many $\sim^b$-equivalence classes B. Hence, if $\succsim$ is complete, this definition also implies that $I \sim J$ iff $I_B \cdot u = J_B \cdot u$ holds for every $\sim^b$-equivalence class B.

A lexitarian representation $\left(\succsim^b, u\right)$ is a rather simple object: it starts with a ranking of positions $\succsim^b$, which can be thought of as "at least as basic as". It also involves a utility function u over positions. The ranking of two profiles I, J by $\left(\succsim^b, u\right)$ is defined by seeking the $\succsim^b$-highest equivalence class that tells them apart, according to their u values. Note that u values over different equivalence classes never get to be compared. Hence we can think of the representation as if there were different utility functions, each defined for a different equivalence class of positions.[10] If it so happens that $\succsim^b$ has finitely many equivalence classes, we can also use the more standard lexicographic representation, with a utility function for each.

We can finally state the main result, proved in Section 6.2:

Theorem 1 *$\succsim$ satisfies A1, A2, A3 and A4 if and only if it has a lexitarian representation $\left(\succsim^b, u\right)$. Moreover, for every $\sim^b$-equivalence class B, u_B is unique up to multiplication by a number $\lambda_B > 0$.*

Observe that the statement of the theorem doesn't specify that $\succsim^b$ is unique. It will be clear from the proof that it is unique when restricting attention to the support of u. Positions that are "null" (i.e., positions p such that $\mathbf{1}_p \sim 0$) can be assigned to various $\sim^b$-equivalence classes.

The statements of the axioms clearly require consideration of populations of varying sizes. However, the implied representation can just as clearly be used to evaluate different allocations for a population of fixed size.

In the case of a finite set of equivalence classes, the representation can take a more familiar form:

Corollary 1 *Assume that $\succsim^b$ has finitely many equivalence classes. Then $\succsim$ satisfies A1, A2, A3 and A4 if and only if there exists $u : \mathcal{P} \to \mathbb{R}_+$ and a partition of $\mathcal{P}$, $(A_1, ..., A_m)$ $(m \geq 1)$ such that $u(p) > 0$ for all $p \notin A_m$, and for all $I, J \in \mathcal{I}$, $I \succsim J$ iff*

$$\left(I_{A_1} \cdot u, ..., I_{A_m} \cdot u\right) \geq_L \left(J_{A_1} \cdot u, ..., J_{A_m} \cdot u\right),$$

where $\geq_L$ is the familiar lexicographic ordering of elements of $\mathbb{R}^m$.

Note that this representation coincides with the lexicographic representation of $\succsim$ by $(u_1, ..., u_m)$ where each u_i vanishes outside A_i. Conversely, given a lexicographic representation by finitely many functions $(u_1, ..., u_m)$ over a population $K = 1, ..., k$ of individuals, their utility profiles can be embedded

[10]For notational simplicity, we use but one function u.

in our model by introducing m equivalence classes of positions, each reflecting the range of u_i for a given i, as explained above for the utilitarian case ($m = 1$). Again, we see that with sufficient flexibility in the formulation of a position, lexicographic person-based and position-based models are equivalent.

A simple and familiar special case is utilitarianism, namely the case in which all positions are commensurate. In that case we may assume a stronger version of the Archimedeanity axiom:

A4* Strong Archimedeanity: For all $I, J, K, L \in \mathcal{I}$, if $K \succ L$, then there exists $n \geq 0$ such that $I + nK \succ J + nL$.

For the statement of this axiom, one need not define commensurability of positions. However, it is obvious that (given the other axioms) any two positions p, q will be commensurate. In other words, given A1-A4, we could also impose the condition that any two positions be commensurate and obtain A4*.

Section 6.1 proves the following:

Theorem 2 $\succsim$ *satisfies A1, A2, and A4* if and only if there exists a function*

$$u : \mathcal{P} \to \mathbb{R}$$

such that, for all $I, J \in \mathcal{I}$

$$\begin{array}{rcl} I & \succsim & J \\ & & \textit{iff} \\ I \cdot u & \geq & J \cdot u \end{array} \tag{3}$$

Moreover, in this case u is unique up to multiplication by a positive number.

Furthermore, if the above hold, then A3 holds iff $u \geq 0$.

Theorem 1 in Chapter 3 of Krantz, Luce, Suppes, and Tversky (1971, p. 74) encompasses the main part of Theorem 2 (i.e., without A3 and nonnegativity) within a more general and abstract framework.[11] Our setting is more concrete,

[11]We thank Jacob Nebel for pointing this out to us. The result is also cited as Proposition 1 in Nebel (2024).

and, as a result, the axioms and proof are commensurately simpler.[12] Indeed, from a mathematical viewpoint, our argument is similar to de Finetti's (1931, 1937) derivation of maximization of expected value relative to a subjective probability (see the version described in Gilboa, 2009). In that model, $\mathcal{P}$ is a set of states of the world, and a real-valued function from the states to the real line describes a bet. Similar axioms are used to derive a separating hyperplane between the bets that are at least as desirable as zero and those that are not, and an additional monotonicity axiom guarantees that (apart from the trivial case of indifference), the gradient of the hyperplane can be interpreted as a probability over the states. The major conceptual difference between the results is that in de Finetti's model the utility function is assumed known (represented by the values of each function), and the derived weights are probabilities—whereas in the present model the given values are numbers of occurrences of positions, and the utility is derived as the weights assigned to these positions. The difference in interpretation also results in a mathematical difference: in the present model the values of the functions compared are only non-negative integers, rather than any real numbers. Consequently, the proof of this result is more involved than it would be in the real-valued case.

In sum, the four axioms are equivalent to the existence of a number $u(p)$ for each position p, such that profiles are evaluated by taking the summation of $u(p)$ over all positions within each equivalence class, and then lexicographically comparing the resulting vectors of sums. The numbers specified by u are unique up to a positive multiplicative constant.[13] There is no assumption

[12] Krantz et al. (1971) deal with an abstract "extensive structure", involving a binary operation that need not be commutative, nor even associative. (An axiom named "weak associativity"guarantees, however, that in case associativity fails, the order of operations does not affect preferences. No such axiom is needed in case commutativity fails.) Thus, one of their axioms isn't needed in our setup, and two others can be simplified. More importantly, the additive operation induces a simple geometric structure, and the proof is basically a separating hyperplane argument. The corresponding proof in Krantz et al. is, naturally, more involved, and relies on representations constructed by "standard sequences" in less structured spaces (including Theorems 4 and 5 in their Chapter 2).

[13] While the set of positions may be infinite, every profile considered has only finitely many individuals, so that the summation of their utilities is well-defined. It is possible, for example, that a position is defined by a level of wealth, and that the set of positions is a

about the utility u as a function of p; the mathematical derivation uses the space of counter vectors (the $I, J, ...$) and not the utility values or even the positions.

3 Modeling Needs by Positions

The formal model above deals with an abstract notion of a "position", which leaves considerable room for interpretation. For example, we may equate a position with a level of income, adopt A4*, and derive standard utilitarianism over income levels, with the derived function $u(p)$ measuring the sympathetic social planner's evaluation of the well-being guaranteed by income p. Alternatively, we may use A1-A4 to derive a ranking of income levels p, and impose an additional condition that no two income levels are commensurate. This (together with monotonicity in income) will derive the leximin criterion (Hammond, 1976). Other axioms could imply that there are only two $\sim^b$-equivalence classes, and that u is constant on each, so that the lexitarian criterion boils down to sufficientarianism (Frankfurt, 1987). Another modeling option would have personal identities encapsulated in positions, so that the position-based model can embed the person-based one.

None of these is the intended interpretation of the model. As explained above, its main goal is to capture need satisfaction levels, ignoring personal identities and delving into determinants of individuals' utilities. How should that be done?

Suppose that there are R needs, and that for each person the extent to which need $r \in \{1, \ldots, R\}$ is met can be described by a number drawn from a finite set N_r. There are several ways in which we may define positions. Let us consider these in the context of a simple example. Suppose that there are two needs, food and space trips, with the former deemed incommensurably more basic than the latter. Each need is classified as unsatisfied or wholly satisfied, so that $N_1 = N_2 = \{0, 1\}$. How should we define $\mathcal{P}$? It seems natural to

continuum. Yet, a given population will only have finitely many individuals, and therefore only finitely many positions p such that $I(p) > 0$.

think of a position as an element of $N_1 \times N_2$, so that, knowing that a person is at a position $p \in N_1 \times N_2$, we know everything about her well-being. With $\mathcal{P} = \{(0,0), (0,1), (1,0), (1,1)\}$, a profile I would indicate that there are $I(0,0)$ people who are starving and make no space trips, $I(1,0)$ people who have enough to eat but don't travel in space, and so on. A lexitarian model $\left(\succsim^b, u\right)$ with a trivial $\succsim^b$ (i.e., a model in which we have $\succsim^b = \mathcal{P} \times \mathcal{P}$ and the strong Archimedean axiom holds) would face the social planner with two choices: either u ignores the second coordinate of the positions, or else one is driven to the conclusion that sufficiently many space trips for some individuals compensate for the starvation of others. To avoid this conclusion, one may wish to use the additional freedom of the lexitarian model. Suppose that $(1,0) \sim^b (1,1) \succ^b (0,0) \sim^b (0,1)$. That is, the relation $\succsim^b$ has two equivalence classes, one in which the individual is well-fed, another in which she starves, with the former ranked strictly above the latter. Within each indifference class, the function u may still represent a preference for space travel, so that we can have $u(1,1) > u(1,0)$ and $u(0,1) > u(0,0)$. However, in this case there will exist a number $n \geq 1$ such that $n(u(1,1) - u(1,0)) > u(1,0)$, meaning that providing space travel to n well-fed individuals can compensate for moving one well-fed individual into the starving $\succsim^b$ equivalence class.

Our focus on needs suggests a different modeling choice: defining $\mathcal{P}$ as the union of N_1, N_2, rather than their cross product. Explicitly, we can have $\mathcal{P} = \{nf, f, ns, s\}$ where "f" and "nf" stand for "food" and "no-food", respectively, and "s" and "ns" are similarly defined for space travel. With a trivial $\succsim^b$ one faces the same problem as above: either $u(ns) = u(s)$, and the social planner ignores preferences for space trips, or else there is a number of space trips that would justify starvation. However, if we have $f \sim^b nf \succ^b s \sim^b ns$, and $u(f) > u(nf)$, $u(s) > u(ns)$, then we obtain a ranking that first considers food, and only when the number of hungry individuals is identical across two profiles, does it compare space trips. This allows the planner to recognize that food and space trips both have value, but to reject trade-offs between the two.

In such a model, each individual is counted exactly once among the positions of a given need. That is, a profile I should satisfy $I(nf) + I(f) =$

$I(ns)+I(s)$ and these equal the total number of individuals in society. However, our model assumes that preferences are defined also for other I's, whose sum across different needs is not constant. Moreover, the definition of commensurability involves indicator profiles (the profiles $\mathbf{1}_p$,$\mathbf{1}_q$ for positions p,q) having 1 at a given position and 0 elsewhere. What type of society do these represent? And will we want to say that $\mathbf{1}_{nf} \succ \mathbf{1}_s$, as required by the definition of $nf \succ^b s$?

In order to cope with this problem, we propose to omit from the model the lowest level of need satisfaction for each need. In this example, we define only two positions: $\mathcal{P} = \{f, s\}$. A profile I now suggests that $I(f)$ individuals are well-fed (and the others have no food), and $I(s)$ enjoy space trips (while the others do not). In this formulation, every profile I has a clear interpretation and it makes sense to consider indicator profiles. Specifically, we will likely have $\mathbf{1}_f \succ \mathbf{1}_s$: a society with one well-fed individual who does not enjoy space trips is preferred to a society with a starving space traveler. Further, we can have $f \succ^b s$ if no number of space trips can compensate for a single person's starvation.

More generally, with R needs, we can define $N'_r = N_r \backslash \{\min N_r\}$ and then consider the union of these as the set of positions: $\mathcal{P} = \bigcup_{r=1}^{R} N'_r$. The planner's preferences over profiles would reveal which needs are considered commensurate to which. For instance, if we add the needs for medication and for opera to the example above, say, $N_3 = N_4 = \{0, 1\}$, the model will have four positions $\mathcal{P} = \{f, s, m, o\}$ (where "m" stands for "having medication" and "o" stands for "opera"). Any profile $I : \mathcal{P} \to \mathbb{Z}_+$ has a natural interpretation, and preferences over these might indicate that $f \sim^b m \succ^b s \sim^b o$, so that food and medication belong to the same equivalence class, and trade-offs between them are conducted in the utilitarian fashion. Similarly, space trips and operas are in an equivalence class, and tradeoffs between them are also dictated by utilitarian addition. But the former are incommensurably more basic than the latter.

Notice that in this model, the social planner will be indifferent to the addition of people whose needs are all at the minimal level. That is, if we add individuals who, for each r, have the r-th need at $\min N_r$, the profile that represents this population is unchanged. Thus, in this model there is no life strictly not worth living, but the very worst life is equivalent to not-living. This interpretation can be used to define the sets N_r: when thinking of a need, we may introduce into the set the lowest conceivable level above which there's some point in living.

Importantly, modeling positions as the union, rather than the cross product, of $\{N_r\}_r$ implies that the social planner thinks of needs in a separably additive way. The additive structure of the model (owing mostly to the Union Axiom) means that the preferences one has over profiles are independent across needs. In a sense, in this formulation of "positions", needs are the carriers of the planner's utility, while individuals are collections of needs. It is as if a social planner who observes a society of m individuals sees m stomachs that need to be filled, m bodies that need to be sheltered, m minds that need to be entertained, and so forth. This separability would typically be satisfied by government policies, if only for administrative reasons: we do not expect a government agency to deprive an individual of public housing because they already received free meals. However, putting administrative difficulties aside, we find that such a separability is also desirable from a normative point of view (for instance because it is consistent with the general principle of incommensurability in needs).

4 Time and Uncertainty

The model as discussed so far implicitly assumes that there is no uncertainty regarding the outcomes of social policies, and that all occur contemporaneously. This is clearly a very restricted setup. People live in different periods of time, and there is considerable uncertainty about the number of people alive at any time, let alone about the position(s) they will occupy. Indeed, some of the most prominent questions of population ethics have to do with the effect of current policies on the uncertainties faced by future generations.

In this section we re-interpret the notion of a "position" to allow us to deal with such questions based on the analysis above. Specifically, we will assume that there is a set of “circumstances”, playing the role of positions in a one-stage model under certainty, and a position will now be a circumstance experienced at a given state of the world and a given period.

Formally, let C be a set of circumstances, let $T = \{0, 1, 2, ...\}$ be a set of time periods, and let Ω be a finite set of states of the world. Assume that $\mathcal{P} = C \times T \times \Omega$. It will be convenient to denote subsets of $\mathcal{I}$ defined by supports that are restricted to states or time periods. With a slight abuse of notation that is unlikely to cause any confusion, we define

$$\begin{aligned}
\mathcal{I}_t &= \{ I \in \mathcal{I} \mid I(c, t', \omega) = 0 \qquad \forall t' \neq t \} \\
\mathcal{I}_\omega &= \{ I \in \mathcal{I} \mid I(c, t, \omega') = 0 \qquad \forall \omega' \neq \omega \} \\
\mathcal{I}_{t,t'} &= \{ I \in \mathcal{I} \mid I(c, t'', \omega) = 0 \qquad \forall t'' \neq t, t' \}
\end{aligned}$$

Define a state ω to be *null* if the value of $I(c, t, \omega)$ does not affect preferences. We impose three additional axioms:

A5 State Invariance: For all non-null $\omega, \omega' \in \Omega$, all $I, J \in \mathcal{I}_\omega$, and $I', J' \in \mathcal{I}_{\omega'}$, if for all $(c, t) \in C \times T$, $I(c, t, \omega) = I'(c, t, \omega')$ and $J(c, t, \omega) = J'(c, t, \omega')$, then $I \succsim J$ iff $I' \succsim J'$.

A6 Time Invariance: For all $t, t' \in T$, all $I, J \in \mathcal{I}_t$, and $I', J' \in \mathcal{I}_{t'}$, if for all $(c, \omega) \in C \times \Omega$, $I(c, t, \omega) = I'(c, t', \omega)$ and $J(c, t, \omega) = J'(c, t', \omega)$, then $I \succsim J$ iff $I' \succsim J'$.

Axioms A5 and A6 are the natural conditions required to have a utility function that is state- and time-independent. They can be stated in other ways as well: first, instead of limiting attention to profiles whose support only involves one state (or one period t), they can be stated for any pairs of profiles I, J, I', J' such that I' and J' are obtained from I and J, respectively, by permuting the corresponding states (or periods). This seemingly stronger requirement will obviously be equivalent to the current formulation in light of the additive structure guaranteed by the results of Section 2.

Clearly, Axioms A5 and A6 consider profiles that do not have the same number of individuals in all states and/or in all time periods. This is in line with our general setup, assuming that profiles with different numbers of individuals can be compared. Indeed, this flexibility of the model seems necessary to discuss some of the problems of population ethics: when considering the long-term risks of climate change or of nuclear wars, the number of individuals alive will typically differ across time periods and across states of the world.

While A5 and A6 treat uncertainty and time symmetrically, the next axiom is specific to the time dimension:

A7 Dynamic Consistency: For all $t \in T$, all $I, J \in \mathcal{I}_{t,t+1}$, and $I', J' \in \mathcal{I}_{t+1,t+2}$, if for all $(c, \omega) \in C \times \Omega$ and $t' \in T$, $I(c, t'+1, \omega) = I'(c, t', \omega)$ and $J(c, t'+1, \omega) = J'(c, t', \omega)$, then $I \succsim J$ iff $I' \succsim J'$.

We start with the (simpler) case of utilitarianism, that is, when all positions are commensurate.

Proposition 1 $\succsim$ *satisfies A1, A2, A4* as well as A5, A6, and A7 if and only if there exists a utility function defined on circumstances,*

$$\hat{u} : C \to \mathbb{R}$$

a probability vector $pr \in \Delta(\Omega)$ and a factor $\delta > 0$ such that, for all $I, J \in \mathcal{I}$

$$\begin{aligned} I &\succsim J \quad \Leftrightarrow \quad V(I) \geq V(J) \qquad (4) \\ &\textit{where} \\ V(I) &= \sum_{\omega \in \Omega} pr(\omega) \sum_{t \in T} \delta^t \sum_{c \in C} I(c, t, \omega)\, \hat{u}(c) \end{aligned}$$

Moreover, in this case $\hat{u}$ is unique up to multiplication by a positive number, whereas pr and δ are unique if $\hat{u}$ is not identically zero.[14]

Corollary 2 *Finally, under the above conditions, A3 holds iff u is nonnegative.*

Note that the proposition does not ensure that $\delta < 1$. Since only vectors I with finite support are considered, there is no mathematical difficulty in

[14] A constant zero-valued u corresponds to the case of a trivial relation for which $I \sim J$ for all I, J.

allowing $\delta \geq 1$. It is not difficult to state an assumption that would guarantee $\delta < 1$.[15]

For the general case of lexitarian preferences, a few adaptations are needed. First, we have to ask which pairs of positions are reasonably commensurate. Our basic intuition is that commensurability is a matter of the circumstances, not of time or uncertainty. For example, luxury goods may be incommensurate to life-saving medication, but there is no reason that luxury goods ten years from now would be incommensurate to the same goods today, or that hunger in the state of global warming would be incommensurate to hunger in the state of no climate change. We therefore impose the following:

A8 Commensurability of time and state: For all $c \in C$, $(t, \omega), (t', \omega') \in T \times \Omega$, (c, t, ω) and (c, t', ω') are commensurate.

We can now state

Proposition 2 $\succsim$ *satisfies A1-A8 iff it has a lexitarian representation* $\left(\succsim^b, u\right)$ *such that:*

(i) There are $\hat{u} : C \to \mathbb{R}_+$, *and, for each* $\sim^b$*- equivalence class* A, *a probability vector* $pr_A \in \Delta(\Omega)$ *and* $\delta_A > 0$ *such that, for all* $(c, t, \omega) \in A$,

$$u(c, t, \omega) = pr_A(\omega)\, \delta_A^t \hat{u}(c)$$

(ii) For every $c \in C$ *and all* $(t, \omega), (t', \omega') \in T \times \Omega$, $(c, t, \omega) \sim^b (c, t', \omega')$.

Moreover, in this case $\hat{u}$ *is unique up to multiplication by a positive number, whereas* pr_A *and* δ_A *are unique if* $\hat{u}$ *is not identically zero.*

Thus, retaining the weaker version of Archimedeanity (A4 rather than A4*), Proposition 2 allows for lexitarian preferences in the more elaborate model, reflecting dynamic and uncertainty considerations. However, axiom A8—clearly reflected in condition (ii) of the representation—restricts the lexicographic nature of preferences to the comparison of basic circumstances.

[15] Such an assumption could compare a vector I in $\mathcal{I}_t$ with the corresponding vector in $\mathcal{I}_{t+1}$—but the direction of preference between them would have to depend on the preference between I and 0.

Hence, other reasons for incommensurabilities, such as lexicographic probabilities as in Blume, Brandenburger, and Dekel (1991), Pivato (2014) or Pivato and Tchouante (2024), are ruled out by A8.

Observe that the axioms, as stated, allow the probability and the discount factor to depend on the equivalence class. One may strengthen A5 and A6 to relate preferences across equivalence classes to obtain a tighter representation where pr_A and δ_A are independent of A.

5 Discussion

This section returns to the three features of public policy formation raised in Section 1.2, explaining how they lead to and are captured by our lexitarian formulation.

5.1 Empathy vs. Sympathy

Theoretical models of utilitarianism typically aggregate a vector of personalized utility functions, specifying the well-being of the various individuals. We are skeptical of such empathetic approaches, for reasons leading us to believe that a sympathetic approach holds out the only hope for applications of utilitarianism.

First, if a planner is to maximize a (weighted) sum of utilities, she must elicit a utility function from each individual. Asking individuals to select a utility function from a set of functions consistent with their true preference rankings is tantamount to asking them how important they think they should be in the social preference. The result is likely to be an arms race: each agent faces unlimited incentives to multiply their utility function by ever-larger constants in an attempt to increase their allocation.

Next, suppose the planner could magically solve this elicitation problem, to the extent that the planner knows the preferences of each individual. Which utility function should the planner now choose to represent these preferences? Harsanyi (1955) is silent on the choice of utility functions, devoting a careful argument to establishing that the planner must maximize *some* weighted sum of utilities while saying nothing about where the weights come from. Dhillon and Mertens (1999) resolve this issue by assuming that every individual's utility function has infimum zero and supremum one, and then invoking an anonymity axiom to assume that every individual receives the same weight in the summation. However, a difficulty with this 0-1 normalization is that individuals can differ in their concepts of "best" and "worst" outcomes. For example, a person of faith who believes in the afterlife can conceive of Heaven and Hell as possible outcomes, which might be beyond the scope of an atheist's comprehension. Requiring the atheist to normalize her mundane best- and worst-cases to the same scale as the believer's effectivelly gives the atheist a higher weight in the social welfare function. Should the atheist promote the maximization of the sum of utilities, she could be thought of as telling the believer, "Well, you look forward to infinite bliss in the afterlife, so why don't you give me all the material goods and live in poverty in the meantime?" This type of reasoning isn't completely foreign to human thought; indeed, believers who join monasteries might be viewed as following such trade-offs between earthly and heavenly payoffs. Yet, it seems problematic to endorse such a policy for a social planner, who would be penalizing the believer for her faith.[16] The planner can escape this conundrum only by imposing some views as to the importance of the members of society, already edging into a sympathetic approach.

Third, empathetic aggregation gives rise to counterintuitive implications even in the simplest of settings. The literature raising and discussing such implications is enormous, and so we present here a single simple example. Assume that each person i's preferences are defined only over their own consumption,

[16] One could instead suggest that the atheist can conceive of Heaven and Hell, but that he attaches zero probabilities to experiencing these outcomes after his death. But this raises the measurability problem again: it might be difficult to design a procedure that would tell us—and the atheist himself—what his utility should be for such extreme outcomes, given that he does not believe any act he takes would result in such outcomes with any positive probability.

with functions

$$
\begin{aligned}
v^1(x^1) &= 2\sqrt{x^1} \\
v^2(x^2) &= x^2 \\
v^3(x^3) &= \frac{1}{2}(x^3)^2.
\end{aligned}
$$

Suppose the planner maximizes the utilitarian criterion $\sum_{i=1}^{3} v^i(x^i)$. Figure 1 illustrates the optimal allocation (x^1, x^2, x^3) as function of the total endowment of resource X.

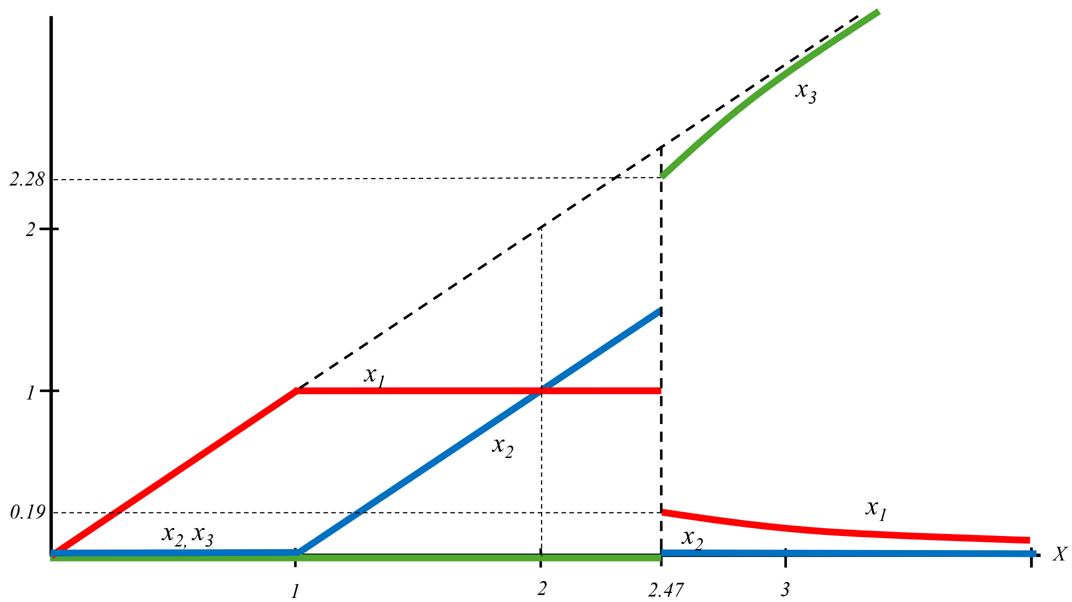

Figure 1: Quantity of resource allocated to person 1 (x_1, red), person 2 (x_2, blue), and person 3 (x_3, green) as a function of total resource X.

Never do all three people receive a positive quantity of resource, and there are both nonmonotonicities and discontinuities of the allocations with respect to the total amount to be shared. Individual 2 initially receives nothing, then experiences a period in which all of any additional resource is given to her, and is then entirely deprived of the resource. Individual 3 initially receives nothing, then jumps to having the largest allocation, and eventually consumes the entire supply. We might seek refuge from this outcome by attaching different weights to the people, but no assignment of weights will resolve all issues.

One may construct an endless variety of such examples. Should individual 1 receive a meager allocation, simply because she is apparently more easily satisfied with smaller rations than either individuals 2 or 3? Should individual 3 receive the bulk of the resource, simply because he is morose and can be placated only by a massive allocation?

If we set out to avoid such quandaries, then the aggregation must not depend on the individual's utilities alone—the aggregation must not be purely empathetic. The planner must step in, at least supplementing the information contained in preferences with the planner's own view on how to aggregate the agents' well being. This brings us to the sympathetic approach.[17]

5.2 Needs vs. Wants

Allowing the planner to invoke her sympathetic assessment of individuals' well-being raises challenges for the planner—how is she supposed to know what makes a person better off? It also raises concerns for society—what keeps the planner from running amok with her discretion? Both difficulties are mitigated by having the planner base her assessment of social welfare on needs rather than wants. As noted in Section 1.2, sympathy and needs are complementary.

It is difficult to specify a sharp boundary between needs and wants. However, we can offer guidelines that make the distinction workable. The considerations that arise in formulating these guidelines reinforce the view that needs rather than wants provide the appropriate foundation for welfare policy.

First, wants are more subjective than needs. Some people fantasize about climbing Mount Everest, others about listening to Wagner's Ring Cycle in Bayreuth. In contrast to these differences, both the former and the latter

[17] Paul Bloom (2016), to whom we owe the "empathy"–"sympathy" nomenclature, argued that sympathy is good for public policy, but empathy is not. A social planner can appropriately understand and make use of people's priorities without adopting their preferences as his own. Adam Smith (1759) and David Hume (1739-40, 18751) have similarly argued for limited empathy.

need food and medication. Wants are hard to identify partly because they are inherently subjective, while there is much less uncertainty (or ambiguity) about needs. This distinction becomes even more pronounced when considering, for example, humanity 200 years hence. What will their wants be? Will they enjoy Beethoven more than techno music? Watching movies or playing computer games? Or are they more likely to enjoy pastimes we can't even conceive of? It seems hard to tell. By contrast, it doesn't seem to be far-fetched to assume that, like us, they will need food and shelter. Subjectivity thus serves as a sign that a certain goal falls into the category of wants rather than of needs.[18]

Second, wants change across periods and cultures more than needs. A significant proportion of the market basket providing the foundation for the consumer price index consists of goods, and the attendant wants, that did not exist a generation ago. In contrast, the needs reflected in the social safety net of developed countries, such as food, housing, education, health and safety, have remained unchanged on a much longer time scale.

Third, and relatedly, wants are elusive because they change with experience. Easterlin's paradox (Easterlin 1973, 1974, Diener, 1984) is based on the idea that one's wants depend on one's aspirations and expectations, which adapt with one's experience (Helson 1947, 1948). This mechanism drove Brickman and Campbell (1971), followed by a large literature, to suggest that there is no way to satisfy wants apart from stepping off the hedonic treadmill.[19] The Sisyphean task of satisfying wants is thus ill-suited as a guideline for welfare policies. However, supplying food and medication is not as hopeless as provid-

[18]The objectivity of needs should be applied at the level of well-being, not goods. For example, not every person needs a wheelchair, and yet we would consider wheelchairs in the category of supplying needs. The relevant need would be "mobility" which is closer to objectivity.

[19]See also Brickman, Coates, and Janoff-Bulman (1978). This rather nihilistic view of well-being has been contested (e.g., Stevenson and Wolfers, 2013), and arguably the only clear conclusion is that research in the social sciences has not reached a conclusion about the definition and measurement of happiness. See Lucas, Dyrenforth, and Diener (2008), Veenhoven and Vergunst (2014), and Easterlin (2017) on Easterlin's paradox and Mancini, Bonanno, and Clark (2011) on the effect of major life events on well-being. See also Kauppinen (2013), Kahneman, Krueger, Schkade, Schwarz, and Stone (2004), Li, Jonah, Wong, and Chao (2019), Schwarz and Clore (1983), and Strack, Martin, and Schwarz (1988) for discussions of ascertaining well-being.

ing expensive jewelry: the needs for clean water or vaccination can be satisfied without requiring ever-increasing levels of water or vaccines.

The common theme appears to be that wants are distinguished from needs by their flexibility: wants change with personal taste, time, culture, and past consumption more than do needs. We thus view needs as a more practicable basis for welfare evaluations. With sufficient empirical research in psychology, we may be able to draw a sharper distinction between needs and wants. It will probably not be perfectly clean, and may end up being a continuum rather than a dichotomy. As a result, some economists may wish to avoid these issues completely, rather than tread the marshes of shady psychological interpretations. However, doing so leaves not only the marshes but also the seas to others who fear no theoretical confusion. We believe there are sufficiently large bodies of clear water out there for economists to introduce these distinctions into their formal models. For instance, it appears evident that owning a private jet isn't a need in the sense that having enough to eat is. It seems equally obvious that social policies exhibit considerably more concern for feeding the hungry than with supplying private jets to the well-nourished.

If a social planner can readily assess needs while remaining clueless about wants, why not avoid questions of classification entirely and think of needs as being whatever the planner cares about? The answer lies, both in principle and in practice, in accountability. Without some (perhaps imperfect) conceptual distinction between needs and wants external to the social planner's decisions, it becomes difficult to assess both the domain and the implied rankings of policy proposals. A conception of needs and wants independent of the behavior of the social planner gives the onlooker the language to point out if the social planner oversteps her mandate, and to object—on moral, not just practical grounds—to poor choices of hierarchies and trade-offs.

The focus on needs rather than wants has the additional advantage of limiting the risks of abuse of the social aggregation rule. We view the specification of needs captured by the function u as reflecting a consensus within the society or polity to which the analysis applies. We envision a person advocating for a particular view of what constitutes a social good as appealing to the axioms in arguing that her position is appropriately couched in terms of the lexitarian aggregation of needs. For support of her vision of what these needs are, she must then appeal to the consensus implicit in the structure and workings of the society around her. One might view the United Nations' Universal Declaration of Human Rights as an attempt to make explicit the social consensus on needs.

One may fear that an appeal to consensus dooms us to the same incompleteness that renders the Pareto criterion an ineffective guide to public policy. However, we note that we do not seek consensus over social policies, but only the set of needs relevant for evaluating those policies. We would hope to reach a wide consensus over questions of the type, "Is food a basic need?" while expecting disagreement over "Should we tax the rich in order to feed the poor?"

While our prime application is for policy decisions at the level of an economy, one can also think of social choice problems in other contexts, where "needs" can be differently interpreted. Consider, for example, parents who have two children, Ada and Bobby. Ada loves mathematics, and obtained a PhD in the field and a tenured academic position. Bobby loves chess. He participates in chess clubs and tournaments, but supports himself by working in an unrelated and unrewarding job. The parents may well consider self-fulfillment as a need, and provide more financial support to Bobby than to Ada, so that Bobby can satisfy the self-fulfillment need as does Ada. In contrast, self-fulfillment may not qualify as a need the social planner takes into account when devising the income tax code.

5.3 Incommensurability

The statement that two needs are incommensurate is a stark one. Is it ever the case that there is absolutely no trade-off between two needs? One may view health as a more important need than enjoyment of the arts, but would we insist that cancer be cured before directing any money to the arts? Given that states invest in education, and sometimes also in the arts, are there any needs that are incommensurate?

The mechanism by which the welfare state satisfies needs is complicated and plagued by uncertainty. Not every dollar appropriated in the health budget is necessarily going to have an effect, let alone to save a life. Hence, even sharp distinctions between needs are blurred by the coarse mechanisms that attempt to satisfy them. The incorporation of uncertainty in Section 4 brings us closer to capturing these complications. We can then conceive of alternatives such as "a society in which disease X is curable with probability Y". It is then no longer the case that one must insist on a 100% curability rate for every disease before one can contemplate funding the arts, while still viewing health as lexicographically more important than artistic fulfillment.

Distinctions between needs are also blurred by instrumental considerations operating over long horizons. When asked, "How many children must be provided with a high-school educations in exchange for the cost of child's death?", one may be tempted to respond that no number of educated children can justify a death. However, if children do not get education at all, the state is unlikely to have scientists in the future. If a cure for a certain type of cancer is to be developed 30 years hence, we must invest in education today. Thus, education is not just an intrinsic need in Maslow's hierarchy: it is also instrumental to other, more basic needs. A state that invests in education can easily justify its budgetary policy by saying that, *in expectation,* and *in the long run*, this investment also saves lives. Following the same line of reasoning, one may try to justify other needs, such as fine arts, instrumentally rather than intrinsically: one can make the claim that, with no artistic and spiritual life, the state might not be able to retain its best talents. A "brain drain" that may result will, eventually, be translated into a reduction in tax collection and to cuts in the budget for basic health needs.

Sharp distinctions between needs are thus elusive. As a result, a model based on such distinctions becomes all the more useful. Public policy invariable confronts contending needs, sometimes viewing them as making comparable claims that need to be balanced and sometimes dismissing some needs as effectively inconsequential in order to focus on more pressing priorities. In a manner that parallels the "inverse optimum" approach of Bourguignon and Spadaro (2012), Jacquet and Lehmann (2021), and Bergstrom and Dodds

(2026), we may ask how to best model the confrontations. A model allowing some needs to be commensurate and others to be incommensurate can be useful in isolating important factors and sharpening questions, thereby refining welfare policies. We view a need-based lexitarian model as vividly capturing the key features of welfare policy formation.

In the final analysis, a model is a simplification of a more complicated world. The utilitarian prescription may appear counterintuitive in its insistence that all needs are commensurate. The lexitarian prescription may also appear counterintuitive in its stark distinction between classes of needs. We believe that in many situations the lexitarian model may be more useful than its utilitarian counterpart.

6 Proofs

The proof of Theorem 1 builds on the proof of Theorem 2, and so we present the latter first.

6.1 Proof of Theorem 2

It is straightforward that the representation (3) implies that $\succsim$ satisfies A1-A3. We thus turn to prove sufficiency of the axioms, from which the uniqueness of the representation will also follow. Naturally, the idea is to extend the relation, given for natural-number vectors, to real-valued vectors and apply a separation theorem. More specifically, we wish to successively extend $\succsim$ to vectors of integer and rational numbers, and then apply a separation theorem to the convex hull of the rational-valued vectors. To this end we define the following sets of vectors:

$$\begin{aligned}\mathcal{I}_{\mathbb{Z}} &= \{\, I : \mathcal{P} \to \mathbb{Z} \mid \#supp\,(I) < \infty \,\} \\ \mathcal{I}_{\mathbb{Q}} &= \{\, I : \mathcal{P} \to \mathbb{Q} \mid \#supp\,(I) < \infty \,\}\end{aligned}$$

For two sets $A \subset B$ and two binary relations on them, $\succsim \subset A \times A$ and $\succsim' \subset B \times B$, we say that $\succsim'$ is an *extension* of $\succsim$ if it extends both the weak and strict preference, that is, if, $\succsim \subset \succsim'$ and $\succ \subset \succ'$. We will also need to refer to the versions of axioms A1-A3 where the vectors are in $\mathcal{I}_{\mathbb{Z}}$ and in $\mathcal{I}_{\mathbb{Q}}$ (rather than only in $\mathcal{I}$). To simplify notation, we will use the same terms "A1", "A2", "A3", rather than redefine them for these domains.

Assume, then, that $\succsim \subset \mathcal{I} \times \mathcal{I}$ satisfies A1-A3. We first extend A2 to apply also for vectors K that are integer-valued but not necessarily nonnegative.

Lemma 1 *There exists a unique $\succsim_{\mathbb{Z}}$ on $\mathcal{I}_{\mathbb{Z}}$ satisfying A1-A3 and extending $\succsim$ (from $\mathcal{I}$ to $\mathcal{I}_{\mathbb{Z}}$).*

Proof: Let there be given $I, J \in \mathcal{I}_{\mathbb{Z}}$. Define $I \succsim_{\mathbb{Z}} J$ if there exists $K \in \mathcal{I}$ such that $I + K \succsim J + K$ (here and in the sequel, the claim that $\succsim$ holds between two vectors is taken to mean that they are both in $\mathcal{I}$ and that the relation holds between them). Clearly, $\succsim_{\mathbb{Z}}$ extends $\succsim$. Completeness of $\succsim$ on $\mathcal{I}$ implies completeness of $\succsim_{\mathbb{Z}}$ on $\mathcal{I}_{\mathbb{Z}}$. To see that $\succsim_{\mathbb{Z}}$ is transitive, assume that for $I, J, L \in \mathcal{I}_{\mathbb{Z}}$ we have $I \succsim_{\mathbb{Z}} J$ and $J \succsim_{\mathbb{Z}} L$. Let $K, K' \in \mathcal{I}$ be such that (i) $I + K \succsim J + K$ and (ii) $J + K' \succsim L + K'$. Applying A2 to (i) and K' and then to (ii) and K, we have $I + K + K' \succsim J + K + K' \succsim L + K + K'$ and thus $I \succsim_{\mathbb{Z}} L$.

Next we wish to show that $\succsim_{\mathbb{Z}}$ satisfies A2 on $\mathcal{I}_{\mathbb{Z}}$. Let there be given $I, J, K \in \mathcal{I}_{\mathbb{Z}}$, with $I \succsim_{\mathbb{Z}} J$. This means that there exists $K' \in \mathcal{I}$ such that $I + K' \succsim J + K'$. Assume first that $K \in \mathcal{I}$. By A2 $I + K' + K \succsim J + K' + K$ so that $I + K \succsim_{\mathbb{Z}} J + K$. Similarly, $I + K \succsim_{\mathbb{Z}} J + K$ implies $I \succsim_{\mathbb{Z}} J$ (using $(K + K') \in \mathcal{I}$).

For a general (not necessarily nonnegative) $K \in \mathcal{I}_{\mathbb{Z}}$, let $K^+, K^- \in \mathcal{I}$ be such that $K = K^+ - K^-$. Then $I + K = I + K^+ - K^-$. Denote $I' = I - K^- \in \mathcal{I}_{\mathbb{Z}}$. Similarly, let $J' = J - K^-$. By the restricted A2 (applied to the nonnegative $K^- \in \mathcal{I}$), $I' \succsim_{\mathbb{Z}} J'$, iff $I' + K^- \succsim_{\mathbb{Z}} J' + K^-$, that is iff $I \succsim_{\mathbb{Z}} J$. Applying the restricted A2 again to I', J', this time with the non-negative $K^+ \in \mathcal{I}$, we get $I' \succsim_{\mathbb{Z}} J'$ iff $I' + K^+ \succsim_{\mathbb{Z}} J' + K^+$. Thus, $I \succsim_{\mathbb{Z}} J$ iff $I' + K^+ \succsim_{\mathbb{Z}} J' + K^+$. It is left to note that $I' + K^+ = I - K^- + K^+ = I + K$ and similarly $J' + K^+ = J + K$, so that $I \succsim_{\mathbb{Z}} J$ iff $I + K \succsim_{\mathbb{Z}} J + K$.

We now turn to A3. Assume that $I, J, K, L \in \mathcal{I}_{\mathbb{Z}}$, and $K \succ_{\mathbb{Z}} L$. Let $K' \in \mathcal{I}$ be such that $I + K', J + K' \in \mathcal{I}$. As $K \succ_{\mathbb{Z}} L$, there exists $K'' \in \mathcal{I}$ such that

$K + K'' \succ L + K''$. Applying A3 (for $\succ$ on $\mathcal{I}$) to $I + K', J + K'$ and the pair $K + K'' \succ L + K''$, we conclude that for some $n \geq 0$ we have $I + K' + n(K + K'') \succ J + K' + n(L + K'')$. This implies $I + nK + (K' + nK'') \succ_{\mathbb{Z}} J + nL + (K' + nK'')$ and, by A2, this is equivalent to $I + nK \succ_{\mathbb{Z}} J + nL$.

By similar arguments, $\succsim_{\mathbb{Z}}$ is the unique extension of $\succsim$ to $\mathcal{I}_{\mathbb{Z}}$ that satisfies the three axioms. $\square$

Lemma 2 *For all $I, J \in \mathcal{I}_{\mathbb{Z}}$, (i) $I \succsim_{\mathbb{Z}} J$ iff $I - J \succsim_{\mathbb{Z}} 0$; (ii) for all $n > 1$, $I \succsim_{\mathbb{Z}} J$ iff $nI \succsim_{\mathbb{Z}} nJ$.*

Proof: Follows from A2 and transitivity. $\square$

We can now make the next step into the rationals.

Lemma 3 *There exists a unique $\succsim_{\mathbb{Q}}$ on $\mathcal{I}_{\mathbb{Q}}$ satisfying A1-A3 and extending $\succsim_{\mathbb{Z}}$ (from $\mathcal{I}_{\mathbb{Z}}$ to $\mathcal{I}_{\mathbb{Q}}$).*

Proof: Let there be given $I, J \in \mathcal{I}_{\mathbb{Q}}$. Define $I \succsim_{\mathbb{Q}} J$ if there exists $n \geq 1$ such that $nI \succsim_{\mathbb{Z}} nJ$. Clearly, $\succsim_{\mathbb{Q}}$ extends $\succsim_{\mathbb{Z}}$. Completeness of $\succsim_{\mathbb{Q}}$ follows from completeness of $\succsim_{\mathbb{Z}}$ on $\mathcal{I}_{\mathbb{Z}}$. To see that transitivity holds as well, assume that $I, J, K \in \mathcal{I}_{\mathbb{Q}}$ with $I \succsim_{\mathbb{Q}} J$ and $J \succsim_{\mathbb{Q}} K$. Let n be such that $nI \succsim_{\mathbb{Z}} nJ$ and let m be such that $mJ \succsim_{\mathbb{Z}} mK$. Applying Lemma 2 to both, we conclude that $nmI \succsim_{\mathbb{Z}} nmJ \succsim_{\mathbb{Z}} nmK$ transitivity of $\succsim_{\mathbb{Z}}$ implies that of $\succsim_{\mathbb{Q}}$.

Next consider A2. It suffices to show that $I \succsim_{\mathbb{Q}} J$ implies $I + K \succsim_{\mathbb{Q}} J + K$ (as the converse argument is symmetric). Let there be given $I, J, K \in \mathcal{I}_{\mathbb{Q}}$, with $I \succsim_{\mathbb{Q}} J$. Let $n \geq 1$ be such that $nI \succsim_{\mathbb{Z}} nJ$. Let $m \geq 1$ satisfy $mK \in \mathcal{I}_{\mathbb{Z}}$. By Lemma 2, $nmI \succsim_{\mathbb{Z}} nmJ$ and A2 yields $nmI + nmK \succsim_{\mathbb{Z}} nmJ + nmK$, that is, $I + K \succsim_{\mathbb{Q}} J + K$.

We now turn to A3. Assume that $I, J, K, L \in \mathcal{I}_{\mathbb{Q}}$, and $K \succ_{\mathbb{Q}} L$. Let $m \geq 1$ satisfy $mJ, mI \in \mathcal{I}_{\mathbb{Z}}$. For some $m' \geq 1$, $m'K \succ_{\mathbb{Z}} m'L$. By Lemma 2, $mm'K \succ_{\mathbb{Z}} mm'L$. Applying A3 (on $\succsim_{\mathbb{Z}}$ on $\mathcal{I}_{\mathbb{Z}}$) to $mm'I, mm'J, mm'K, mm'L$ we infer that there exists $n \geq 0$ such that $mm'I + nmm'K \succ_{\mathbb{Z}} mm'J + nmm'L$, which, by definition, means that $I + nK \succ_{\mathbb{Q}} J + nL$.

Finally, similar arguments imply that the extension $\succsim_{\mathbb{Q}}$ is unique. $\square$

Lemma 4 *For all $I, J \in \mathcal{I}_{\mathbb{Q}}$, (i) $I \succsim_{\mathbb{Q}} J \iff I - J \succsim_{\mathbb{Q}} 0$; (ii) for all $n > 1$, $I \succsim_{\mathbb{Q}} J \iff nI \succsim_{\mathbb{Q}} nJ$.*

Proof: Identical to the proof of Lemma 2. $\square$

It is worth mentioning explicitly the following:

Lemma 5 *For all $I, J \in \mathcal{I}_{\mathbb{Q}}$, and every rational number $\alpha > 0$, $I \succsim_{\mathbb{Q}} J$ iff $\alpha I \succsim_{\mathbb{Q}} \alpha J$ (and thus $I \succ_{\mathbb{Q}} J$ iff $\alpha I \succ_{\mathbb{Q}} \alpha J$).*

Proof: Follows from Lemma 4. $\square$

We turn to the definition of the function u. If $I \sim J$ for all $I, J \in \mathcal{I}$, select $u(\cdot) \equiv 0$. Otherwise, assume that $I^* \succ J^*$ for some $I^*, J^* \in \mathcal{I}$. Consider a finite $\mathcal{P}_0 \subset \mathcal{P}$ such that $\mathcal{P}^* \equiv \mathit{supp}(I^*) \cup \mathit{supp}(J^*) \subset \mathcal{P}_0$. We will define u for such a subset $\mathcal{P}_0$ and then show how these definitions can be "patched" together to a definition of u on all of $\mathcal{P}$. Let $\mathcal{I}_{\mathbb{Q}_0}$ be the set of vectors in $\mathcal{I}_{\mathbb{Q}}$ that vanish outside of $\mathcal{P}_0$. Define

$$\begin{aligned} A &= \{ I \in \mathcal{I}_{\mathbb{Q}_0} \mid I \succ_{\mathbb{Q}} 0 \} \\ B &= \{ I \in \mathcal{I}_{\mathbb{Q}_0} \mid 0 \succ_{\mathbb{Q}} I \} \\ E &= \{ I \in \mathcal{I}_{\mathbb{Q}_0} \mid I \sim_{\mathbb{Q}} 0 \} \end{aligned}$$

We consider A, B, E as subsets of the finite-dimensional Euclidean space $\mathbb{R}^{|\mathcal{P}_0|}$, and observe the following: (I) A, B, E are pairwise disjoint; (II) $A \cup B \cup E = \mathcal{I}_{\mathbb{Q}_0}$; (III) $A = -B$; (IV) $0 \in E$; (V) for all $I, J \in \mathcal{I}_{\mathbb{Q}_0}$, $I \succ_{\mathbb{Q}} J \iff I - J \in A$, $J \succ_{\mathbb{Q}} I \iff I - J \in B$, and $I \sim_{\mathbb{Q}} J \iff I - J \in E$; and (VI) $A, B \neq \varnothing$.

Lemma 6 *Suppose that $I, J \in A(B, E)$ and $\alpha \in [0,1] \cap \mathbb{Q}$. Then $\alpha I + (1-\alpha) J \in A(B, E)$.*

Proof: Follows from Lemma 5, A2, and transitivity. $\square$

In the following, for $I \in \mathbb{R}^{|\mathcal{P}_0|}$ and $\varepsilon > 0$, $N_\varepsilon(I)$ denotes the ε-neighborhood of I in $\mathbb{R}^{|\mathcal{P}_0|}$.

Lemma 7 *Let $I \in A(B)$. There exists $\varepsilon > 0$ such that for every $J \in N_\varepsilon(I) \cap \mathcal{I}_{\mathbb{Q}_0}$, we have $J \in A(B)$.*

Proof: Let there be given $I \in A$. For $1 \leq j \leq |\mathcal{P}_0|$, let e_j be the j-th unit vector in $\mathbb{R}^{|\mathcal{P}_0|}$. We wish to find an $\varepsilon_j > 0$ such that $[I - \varepsilon_j e_j, I + \varepsilon_j e_j] \cap \mathcal{I}_{\mathbb{Q}_0} \subset A$. Consider $K^+ = I + e_j \in \mathcal{I}_{\mathbb{Q}_0}$. By A3 there exists $n \geq 1$ such that $K^+ + nI \succ_{\mathbb{Q}} 0$. Hence by Lemma 4), $\frac{1}{n+1}[K^+ + nI] \succ_{\mathbb{Q}} 0$. Thus, for $\varepsilon_j^+ = \frac{1}{n+1}$ we have $\left[I, I + \varepsilon_j^+ e_j\right] \cap \mathcal{I}_{\mathbb{Q}_0} \subset A$. Similarly, there exists $\varepsilon_j^- > 0$ such that $\left[I - \varepsilon_j^- e_j, I\right] \cap \mathcal{I}_{\mathbb{Q}_0} \subset A$—and we define $\varepsilon_j = \min\left(\varepsilon_j^+, \varepsilon_j^-\right)$.

Let $\bar{\varepsilon} = \min_j \varepsilon_j$ and set $\varepsilon = \frac{\bar{\varepsilon}}{\sqrt{|\mathcal{P}_0|}} > 0$. It follows that the ε-ball around I is included in the convex hull of $\{I - \bar{\varepsilon} e_j, I + \bar{\varepsilon} e_j\}_{j \leq |\mathcal{P}_0|}$. Consider a rational point $J \in N_\varepsilon(I) \cap \mathcal{I}_{\mathbb{Q}_0}$. By Carathëodory's Theorem, there are $l \leq |\mathcal{P}_0| + 1$ points $\{I_r\}_{r \leq l}$ in $\{I - \bar{\varepsilon} e_j, I + \bar{\varepsilon} e_j\}_{j \leq |\mathcal{P}_0|}$ and nonnegative numbers $\{\alpha_r\}_{r \leq l}$ adding up to 1 such that $J = \sum_{r \leq l} \alpha_r I_r$, and, furthermore, these $\{\alpha_r\}_{r \leq l}$ are unique (for the given set $\{I_r\}_{r \leq l}$). This implies that $\{\alpha_r\}_{r \leq l}$ are rational (as the unique solution to a system of equations with rational numbers). It follows, from iterated applications of of Lemma 6, that $J = \sum_{r \leq l} \alpha_r I_r \in A$.

The argument for B is symmetric. $\square$

Lemma 8 *For every $I \in E$ there are arbitrarily close points $J^+ \in A$ and $J^- \in B$.*

Proof: Follows from Lemma 4, A2, and transitivity. $\square$

Lemma 9 *There exists $u : \mathcal{P}_0 \to \mathbb{R}$ such that, for all $I, J \in \mathcal{I}_{\mathbb{Q}_0}$*

$$\begin{array}{rcl} I & \succsim & {}_{\mathbb{Q}_0} J \\ & & \textit{iff} \\ \sum_{p \in \mathcal{P}_0} I(p)\, u(p) & \geq & \sum_{p \in P_0} J(p)\, u(p) \end{array}$$

Further, u is unique up to multiplication by a positive number.

Proof: We know that A and B are closed under rational convex combinations. Consider $A^0 = \mathit{int}(\mathit{conv}(A))$ and $B^0 = \mathit{int}(\mathit{conv}(B))$, where *conv* refers to the convex hull in $\mathbb{R}^{|\mathcal{P}_0|}$ (including not-necessarily rational points, and not-necessarily rational convex combinations). By Lemma 7 the sets $A^0, B^0 \subset \mathbb{R}^{|\mathcal{P}_0|}$, which are open and convex, are non-empty (and full-dimensional). By Lemma 7 we also know that $A \subset A^0$ and $B \subset B^0$. Moreover, for any $I \in A^0(B^0)$ there exists $\varepsilon > 0$ such that $N_\varepsilon(I) \cap \mathcal{I}_{\mathbb{Q}_0} \subset A(B)$. In particular, this means that $A^0 \cap B^0 = \varnothing$: if there were $x \in A^0 \cap B^0$, we would be able to find two neighborhoods thereof, such that in one all rational points have to be in A and in the other—in B, in contradiction to $A \cap B = \varnothing$.

Thus we can apply a separating hyperplane theorem to conclude that there exists $u : \mathcal{P}_0 \to \mathbb{R}$ and a number $c \in \mathbb{R}$ such that

$$
\begin{array}{llll}
I \in & A^0 & \Longrightarrow & I \cdot u > c \qquad \text{(i)} \\
I \in & B^0 & \Longrightarrow & I \cdot u < c \qquad \text{(ii)}
\end{array}
$$

We first note that the implications in (i) and in (ii) can also be reversed. To see this, assume that $I \cdot u > c$. Choose $\varepsilon > 0$ such that for every $J \in N_\varepsilon(I)$, we have $J \cdot u > c$. We argue that, if $J \in \mathcal{I}_{\mathbb{Q}_0}$, we have $J \in A$. Indeed, if $J \in B$ then $J \in B^0$ and (ii) would imply $J \cdot u < c$. If, however, $J \in E$, then, by Lemma 8, there are points $J' \in B$ arbitrarily close to J, so that we can pick $J' \in N_\varepsilon(I) \cap B$. This would contradict (ii) again. Thus all rational points in $N_\varepsilon(I)$ are in A and $I \in A^0$ follows. The argument is symmetric for B (and the converse of (ii)).

Next, observe that for $I \in E$ we have to have $I \cdot u = c$, because an inequality would imply that I is in A or in B. Finally we observe that $c = 0$ because $0 \in E$.

We conclude that for all $I \in \mathcal{I}_{\mathbb{Q}}$

$$
\begin{array}{lll}
I \in A & \Longleftrightarrow & I \cdot u > 0 \\
I \in E & \Longleftrightarrow & I \cdot u = 0 \\
I \in B & \Longleftrightarrow & I \cdot u < 0
\end{array}
$$

which means that, for all $I, J \in \mathcal{I}_{\mathbb{Q}_0}$, $I \succsim_{\mathbb{Q}_0} J$ ($I \succ_{\mathbb{Q}_0} J$) iff $I - J \in A \cup E$ ($I - J \in A$) iff $(I - J) \cdot u \geq 0$ (> 0)—and u is unique up to multiplication by a positive constant $\lambda > 0$. $\square$

To conclude the proof of the theorem, we now define u for all p. We apply Lemma 9 and fix one $u_0 : \mathcal{P}_0 \to \mathbb{R}$ that represents $\succsim_{\mathbb{Q}_0}$ on $\mathcal{I}_{\mathbb{Q}_0}$. For every $p \notin \mathcal{P}_0$, we consider $\mathcal{P}_p = \mathcal{P} \cup \{p\}$ and the corresponding $\mathcal{I}_{\mathbb{Q}_p}$. There exists a unique function u_p that extends u_0 to $\mathcal{P}_p$ and represents $\succsim_{\mathbb{Q}_p}$. Let $u(p) = u_p(p)$. Thus u is defined for all of $\mathcal{P}$.

To see that u so defined represents $\succsim_{\mathbb{Q}}$ on all of $\mathcal{I}_{\mathbb{Q}}$, consider $K, L \in \mathcal{I}_{\mathbb{Q}}$. Limit attention to $\mathcal{P}' = \mathcal{P}^* \cup supp(K) \cup supp(L)$. By the same arguments, there exists a $u' : \mathcal{P}' \to \mathbb{R}$ that represents $\succsim_{\mathbb{Q}'}$. It thus also represents $\succsim_{\mathbb{Q}_0}$ on $\mathcal{I}_{\mathbb{Q}_0}$ and $\succsim_{\mathbb{Q}_p}$ on $\mathcal{I}_{\mathbb{Q}_p}$ for every $p \in \mathcal{P}'$. Hence it has to equal u up to multiplication by a positive constant. $\square$

6.2 Proof of Theorem 1

We begin with sufficiency of the axioms. We first note that

Lemma 10 *For all $I, J, K, L \in \mathcal{I}$ and all $m \geq 1$,*
(i) $I \succsim J$ and $K \succsim (\succ) L$ imply $I + K \succsim (\succ) J + L$;
(ii) If $K \sim L$ then $I \succsim J$ iff $I + K \succsim J + L$;
(iii) $I \succsim (\succ) J$ implies $mI \succsim (\succ) mJ$.

Proof: (i) follows from two applications of A2 and transitivity: $I \succsim J$ iff $I + K \succsim J + K$ and $K \succsim L$ iff $J + K \succsim J + L$.

(ii) follows from two applications of (i) if $I \succsim J$ then $K \succsim L$ implies $I + K \succsim J + L$. Conversely, if $I + K \succsim J + L$, had $J \succ I$ been the case, (i) would yield $J + L \succ I + K$, a contradiction.

(iii) follows from iterated applications of (i). $\square$

We now turn to define the relation $\succsim \cdot$: For $p, q \in \mathcal{P}$, $p \succsim \cdot q$ if $\exists k \geq 1$ such that $k\mathbf{1}_p \succsim \mathbf{1}_q$.

Lemma 11 *(i) $\succsim \cdot$ is complete;*
(ii) $\succsim \cdot$ is transitive;
(iii) $\sim \cdot$ coincides with commensurability.

Proof: (i) follows from completeness of $\succsim$, say, applied to $\mathbf{1}_p$ and $\mathbf{1}_q$.

(ii) follows from transitivity of $\succsim$ and Lemma 10: assume that $p \succsim \cdot q$ and $q \succsim \cdot r$. Let $k, l \geq 1$ be such that $k\mathbf{1}_p \succsim \mathbf{1}_q$ and $l\mathbf{1}_q \succsim \mathbf{1}_r$. Part (iii) of Lemma 10 implies that $kl\mathbf{1}_p \succsim l\mathbf{1}_q$ and transitivity—that $kl\mathbf{1}_p \succsim \mathbf{1}_r$, hence $p \succsim \cdot r$.

Finally, (iii) is immediate. □

Let us define the $\sim \cdot$-equivalence classes to be $\mathcal{A} \equiv \mathcal{P}/\sim \cdot$. We use $\succsim \cdot$ over $\mathcal{A}$ with the obvious meaning. We can write, for each I,

$$I = \sum_{A \in \mathcal{A}} I_A$$

and note that $I_A = 0$ for all but finitely many A's. Equivalently, for every I there is a finite set of $\sim \cdot$-equivalence classes, $(A_1, ..., A_m)$ such that $A_i \succ \cdot A_{i+1}$ (for $1 \leq i < m$) and

$$I = \sum_{i \leq m} I_{A_i}$$

(and $I_B = 0$ for all $B \in \mathcal{A} \setminus \{A_1, ..., A_m\}$).

The key step in the proof is to show that, if two profiles I, J can be ranked on a given $A \in \mathcal{A}$, anything that happens in lower indifference classes of positions does not matter. Formally,

Lemma 12 *Let there be given $I, J \in \mathcal{I}$ and $A \in \mathcal{A}$ such that*

(i) $I_A \succ J_A$

(ii) $I_B \sim J_B$ for all $B \in \mathcal{A}$ with $B \succ \cdot A$;

Then $I \succ J$.

Proof: We first write

$$\begin{aligned} I &= \sum_{\substack{B \in \mathcal{A} \\ B \succ \cdot A}} I_B + I_A + \sum_{\substack{C \in \mathcal{A} \\ A \succ \cdot C}} I_C \\ J &= \sum_{\substack{B \in \mathcal{A} \\ B \succ \cdot A}} J_B + J_A + \sum_{\substack{C \in \mathcal{A} \\ A \succ \cdot C}} J_C \end{aligned}$$

and it will be useful to look also at

$$
\begin{aligned}
I^* &= \sum_{\substack{B\in\mathcal{A}\\ B\succ\cdot A}} I_B + I_A \\
J^* &= \sum_{\substack{B\in\mathcal{A}\\ B\succ\cdot A}} J_B + J_A
\end{aligned}
$$

Because $I_A \succ J_A$, we can Lemma 10 consecutively for every $B \succ \cdot A$ such that $I_B \neq 0$ or $J_B \neq 0$ (and there are finitely many such B's) to conclude that $I^* \succ J^*$. We now wish to prove, that we can change I^* and J^* on any position p in C with $A \succ \cdot C$, without changing the strict preference. As there are finitely many positions that distinguish (I, J) from (I^*, J^*), the argument can be used inductively to show that $I^* \succ J^*$ does indeed imply $I \succ J$.

Let there be given such a p, so that $q \succ \cdot p$ for every $q \in A$. Consider the modified profiles

$$
\begin{aligned}
I_p^* &= I^* + I(p)\,\mathbf{1}_p \\
J_p^* &= J^* + J(p)\,\mathbf{1}_p
\end{aligned}
$$

We wish to show that $I_p^* \succ J_p^*$. We will use A4 to conclude that $I_p^* \succsim I^*$, and thus it would suffice to prove that $I^* \succ J_p^*$.[20] Assume not. Then

$$J_p^* = J^* + J(p)\,\mathbf{1}_p \succsim I^*$$

Consider any $q \in A$. Apply the Archimedean axioms to the four profiles $0, \mathbf{1}_q, I_A, J_A$ whose supports are all included in A. Since $I_A \succ J_A$, there exists $k \geq 1$ such that $(0+)\, kI_A \succ \mathbf{1}_q + kJ_A$. (That is, adding one at position q does not suffice to reverse the strict preference $kI_A \succ kJ_A$.) However, $J^* + J(p)\,\mathbf{1}_p \succsim I^*$ implies (by Lemma 10 (iii)) that

$$kJ^* + kJ(p)\,\mathbf{1}_p \succsim kI^*$$

Using Lemma 10 (ii), when applied to all B with $B \succ \cdot A$ (for which $I_B \sim J_B$ and $kI_B \sim kJ_B$), we also have

$$kJ_A + kJ(p)\,\mathbf{1}_p \succsim kI_A \succ \mathbf{1}_q + kJ_A$$

[20] If A4 is not assumed, the argument is similar.

The preference $kJ_A + kJ(p)\mathbf{1}_p \succ \mathbf{1}_q + kJ_A$ implies, using Lemma 10 (i), $kJ(p)\mathbf{1}_p \succ \mathbf{1}_q$ which means that $p \succsim \cdot q$—a contradiction.

Thus we have $I_p^* \succ J_p^*$ and, by repeating the argument inductively for every p in $supp(I) \cup supp(J)$ with $q \succ \cdot p$ for $q \in A$, we conclude that $I \succ J$. $\square$

We now complete the proof of the theorem. For every $A \in \mathcal{A}$, we restrict attention to

$$\mathcal{I}_A = \{ I_A \mid I \in \mathcal{I} \}$$

and notice that $\succsim$ satisfies A1-A4 on this space, where A3, the unrestricted version of Archimedeanity, follows from the fact that all I_A vanish outside of the equivalence class A. Theorem 2 guarantees that existence of a nonnegative u_A that represents $\succsim$ on $\mathcal{I}_A$ via the product operation. The function u has to vanish outside of A. We define $u = \sum\limits_{A \in \mathcal{A}} u_A$ which is well-defined because each p belongs to a unique A.[21] It is straightforward to verify that $(\succsim \cdot, u)$ is a lexitarian representation of $\succsim$.

The proof of necessity of the axioms is straightforward and therefore omitted.

Clearly, we obtain a function u that can be separately re-scaled on each $A \in \mathcal{A}$, but is otherwise unique: there can be at most one $A \in \mathcal{A}$ consisting of p with $u(p) = 0$, where for $q \in A^c$ we have $u(q) > 0$. Further, for $q_1, q_2 \in A^c$, $u(q_1)/u(q_2)$ is uniquely defined. If we wish the relation $\succsim \cdot$ to have maximal equivalence classes, as in our construction, then it is unique. However, one may have an equivalent representation with other relations $\succsim' \cdot$, which agree on the equivalence classes over which $u(q) > 0$ but may not consider all positions p with $u(p) = 0$ to be $\sim' \cdot$ equivalent. $\square$

6.3 Proof of Proposition 1

It is straightforward to see that the axioms are necessary: given a representation of $\succsim$ by $V(I) = \sum_{\omega \in \Omega} pr(\omega) \sum_{t \in T} \delta^t \sum_{c \in C} I(c, t, \omega)\hat{u}(c)$, we have a representation as in Theorem 2 with $u(c, t, \omega) = pr(\omega)\delta^t \hat{u}(c)$. This means

[21] Alternatively, one could consider the restrictions of profiles to equivalence classes, obtain a function u_A that is defined only over A , and define u as their union.

that, restricting attention to profiles in $\mathcal{I}_\omega$ or in $\mathcal{I}_{\omega'}$ preferences are identical if both ω, ω' are non-null (or if both are null, which means that $u(c,t,\omega) = u(c,t,\omega') = 0$ for all $(c,t) \in C \times T$). Similarly, A6 and A7 follow from the representation.

Conversely, assume that A5-A7 hold. Consider non-null $\omega, \omega' \in \Omega$ and their corresponding $\mathcal{I}_\omega$,$\mathcal{I}_{\omega'}$. Applying Theorem 2 to each of $\mathcal{I}_\omega$,$\mathcal{I}_{\omega'}$ separately, we conclude that there exist functions $u_\omega, u_{\omega'} : C \times T \to \mathbb{R}$ that represent $\succsim$ on these restricted spaces via the inner products (i.e., for $I, J \in \mathcal{I}_\omega$, $I \succsim J$ iff $I \cdot u_\omega \geq J \cdot u_\omega$, and similarly for $I, J \in \mathcal{I}_{\omega'}$). By A5, $\succsim$ on $\mathcal{I}_\omega$ is identical to its restriction to $\mathcal{I}_{\omega'}$. Thus, u_ω also represents $\succsim$ on $\mathcal{I}_{\omega'}$ (and vice versa). By the uniqueness result, $u_{\omega'}$ is a positive multiple of u_ω. Fix one non-null state ω_0 (if such exists) and for every ω' choose $\lambda_{\omega'} > 0$ such that $u_{\omega'} = \lambda_{\omega'} u_{\omega_0}$. That is, $u(c,t,\omega') = \lambda_{\omega'} u(c,t,\omega_0)$. If all states are null, u vanishes. Otherwise, normalize $(\lambda_{\omega'})_{\omega'}$ to be a probability vector pr.

Next consider two periods $t, t' \in T$ and, using similar reasoning applied to A6, conclude that, for every $t > 0$ there exists $\mu_t > 0$ such that $u(c,t,\omega_0) = \mu_t u(c,0,\omega_0)$. Define $\hat{u}(c) = u(c,0,\omega_0)$ and observe that $u(c,t,\omega) = pr(\omega)\,\mu_t \hat{u}(c)$. Finally, invoke A7 to conclude that, for $\delta \equiv \mu_1 > 0$ we have to have $\mu_t = \delta^t$ for all t. $\square$

6.4 Proof of Proposition 2

We start with necessity of the axioms. A8 follows from condition (ii). We first note that, for each $\sim^b$-equivalence class A, A5-A7 hold for profiles whose supports are contained in it, $\mathcal{I}_A$. This follows from the necessity of the axioms in Proposition 1. To see that these axioms hold in general, it remains to follow the lexicographic representation. For example, consider A5. Let there be given two non-null $\omega, \omega' \in \Omega$ and $I, J \in \mathcal{I}_\omega$, and $I', J' \in \mathcal{I}_{\omega'}$ with $I(c,t,\omega) = I'(c,t,\omega')$ and $J(c,t,\omega) = J'(c,t,\omega')$ for all $(c,t) \in C \times T$. We need to show that $I \succsim J$ iff $I' \succsim J'$. Assume that $I \succ J$. Let A be the $\succsim^b$-maximal equivalence class such that $I_A \nsim J_A$ (that is, $I_A \succ J_A$). Then, for any equivalence class $A' \succ^b A$ we have $I_{A'} \sim J_{A'}$ and hence $I'_{A'} \sim J'_{A'}$. Similarly, for A we have $I'_A \succ J'_A$ and thus $I' \succ J'$. If, however, $I \sim J$, we have to have

$I_A \sim J_A$ for all A and thus also $I'_A \sim J'_A$ and $I \sim J$ follows. Similar reasoning applied to the necessity of A6 and A7.

To prove sufficiency of the axioms, we apply the sufficiency part of Proposition 1 for each $\sim^b$-equivalence class A separately, and derive the representation as in (i) with $u(c,t,\omega) = pr_A(\omega)\,\delta_A^t \hat{u}(c)$ for all (c,t,ω) (where pr_A and δ_A are derived from Proposition 1). Finally, (ii) follows from A8. $\square$

7 Bibliography

Adler, Matthew D. (2012), Well-Being and Fair Distribution: Beyond Cost-Benefit Analysis. Oxford University Press.

Alcantud, José Carlos R., Marco Mariotti, and Roberto Veneziani (2022), "Sufficientarianism", *Theoretical Economics*, **17**: 1529-1557.

Arrhenius, Gustaf (2000), "An Impossibility Theorem for Welfarist Axiologies", Economics and Philosophy, 16: 247-266.

Asheim, Geir B. and Susumu Cato (2026), "The Dilemma between Repugnancy and Sadism Extends beyond Critical-Level Utilitarianism", Economics and Philosophy, forthcoming.

Bentham, Jeremy (1780/1789), *An Introduction to the Principles of Morals and Legislation*. London: T. Payne and Son.

Bergstrom, Katy and William Dodds (2026), "A General Theory of Inverse Welfare Functions", *Theoretical Economics*, **21**: 973-1018.

Blackorby, Charles and David Donaldson (1984), "Social Criteria for Evaluation Population Change", Journal of Public Economics, 25: 13-33.

Blackorby, Charles, Walter Bossert, and David Donaldson (1995), "Intertemporal Population Ethics: Critical-Level Utilitarian Principles", *Econometrica*, **65**: 1303-1320.

Blackorby, Charles, Walter Bossert, and David Donaldson (1997), "Critical-Level Utilitarianism and the Population Ethics Dilemma", *Economics and Philosophy*, **13**: 197-230.

Bloom, Paul (2016), *Against Empathy: The Case for Rational Compassion*. Ecco/Harper Collins.

Blume, Lawrence, Adam Brandenburger, and Eddie Dekel (1991), "Lexicographic Probabilities and Choice under Uncertainty", *Econometrica*, **59**: 61-79.

Boonin, David (2014), The Non-Identity Problem and the Ethics of Future People. Oxford University Press.

Bossert, Walter (1990), "Maximin Welfare Orderings with Variable Population Size", Social Choice and Welfare, 7: 39-45.

Bossert, Walter and Kohei Kamaga (2020), "An Axiomatization of the Mixed Utilitarian-Maximin Social Welfare Orderings", *Economic Theory*, **69**: 451-473.

Bossert, Walter, Susumu Cato, and Kohei Kamaga (2023), "Thresholds, Critical Levels, and Generalized Sufficientarian Principles", *Economic Theory*, **75**: 1099-1139.

Bourguignon, François, and Amedeo Spadaro (2012), "Tax-Benefit Revealed Social Preferences", *Journal of Economic Theory*, **147**: 2134-2157.

Brickman, Philip, and Donald T. Campbell (1971), "Hedonic Relativism and Planning the Good Society", in M. H. Appley (Ed.), *Adaptation Level Theory: A Symposium.* Academic Press.

Brickman, Philip, Dan Coates, and Ronnie Janoff-Bulman (1978), "Lottery Winners and Accident Victims: Is Happiness Relative?", *Journal of Personality and Social Psychology*, **36**: 917-927.

Broome, John (2004), *Weighing Lives.* Oxford University Press.

Carlson, Erik (1998), "Mere Addition and the Two Trilemmas of Population Ethics", Economics and Philosophy, 14: 283-306.

Carlson, Erik (2010), "Generalized Extensive Measurement for Lexicographic Orders", *Journal of Mathematical Psychology*, **54**: 345-351.

Cooper, John M. (1998), "Pleasure and Desire in Epicurus", in *Reason and Emotion: Essays on Ancient Moral Psychology and Ethical Theory.* Princeton: Princeton University Press: 485-514.

de Finetti, Bruno (1931), "Sul Significato Soggettivo della Probabilitá", *Fundamenta Mathematicae*, **17**: 298-329.

de Finetti, Bruno (1937), "La Prévision: Ses Lois Logiques, Ses Sources

Subjectives", *Annales de l'Institut Henri Poincaré*, **7**: 1-68.

Dhillon, Amrita, and Jean-François Mertens (1999), "Relative Utilitarianism", *Econometrica*, **67**: 471-498.

Diener, Ed (1984), "Subjective Well-being", *Psychological Bulletin*, **95**: 542-575.

Easterlin, Richard A. (1973), "Does Money Buy Happiness?", *The Public Interest*, **30**: 3-10.

Easterlin, Richard A. (1974), "Does Economic Growth Improve the Human Lot? Some Empirical Evidence", in *Nations and Households in Economic Growth.* P. A. David and M. W. Reder (Eds.), Academic Press: 89-125.

Easterlin, Richard A. (2017), "Paradox Lost?", *Review of Behavioral Economics*, **4**: 311-339.

Echenique, Federico, and Quitzé Valenzuela-Stookey (2024), "Utilitarian Social Choice and Distributional Welfare Analysis", arXiv/UC Berkeley.

Elliot, Andrew J. and Martin V. Covington (2001), "Approach and Avoidance Motivation", *Educational Psychology Review*, **13**: 73-92. https://doi.org/10.1023/A:100900901

Fontaine, Philippe (1997) "Identification and Economic Behavior: Sympathy and Empathy in Historical Perspective", *Economics and Philosophy*, **13**: 261-280.

Fontaine, Philippe (2001), "The Changing Place of Empathy in Welfare Economics", *History of Political Economy*, **33**: 387-410.

Frankfurt, Harry G. (1987), "Equality as a Moral Ideal", *Ethics*, **98**: 21-43.

Frick, Johann (2020), "Conditional Reasons and the Procreation Asymmetry", Philosophical Perspectives, 34: 53-87.

Fryxell, Loren K. (2024), "XU" (Experienced Utility), working paper.

Gilboa, Itzhak (2009), Theory of Decision under Uncertainty. Econometric Society Monograph Series No. 45. Cambridge: Cambridge University Press.

Gustafsson, Johan E. (2022), "Our Intuitive Grasp of the Repugnant Conclusion", in The Oxford Handbook of Population Ethics: 371-389.

Hammond, Peter J. (1976), "Equity, Arrow's Conditions, and Rawls' Difference Principle", *Econometrica*, **44**: 793-804.

Harsanyi, John C. (1953), "Cardinal Utility in Welfare Economics and in the Theory of Risk-Taking", *Journal of Political Economy*, **61**: 434-435.

Harsanyi, John C. (1955), "Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparisons of Utility", *Journal of Political Economy*, **63**: 309-321.

Helson, Harry (1947), "Adaptation-Level as Frame of Reference for Prediction of Psychophysical Data", *American Journal of Psychology*, **60**: 1-29.

Helson, Harry (1948), "Adaptation-Level as a Basis for a Quantitative Theory of Frames of Reference", *Psychological Review*, **55**: 297-313.

Huemer, Michael (2008), "In Defence of Repugnance", Mind, 117(468): 899-933.

Hume, David (1739-40), *A Treatise of Human Nature.* Edited by L. A. Selby-Bigge and P. H. Nidditch. Oxford: Clarendon Press, 1978.

Hume, David (1751), *An Enquiry Concerning the Principles of Morals.* Edited by Tom L. Beauchamp. Oxford: Oxford University Press, 1998.

Hurka, Thomas (1983), "Value and Population Size", Ethics, 93: 496-507.

Jacquet, Laurence, and Etienne Lehmann (2021), "Optimal Income Taxation with Composition Effects", Journal of the European Economic Association, 19: 1299-1341.

Kahneman, Daniel, Alan B. Krueger, David A. Schkade, Norbert Schwarz, and Arthur A. Stone (2004), "A Survey Method for Characterizing Daily Life Experience: The Day Reconstruction Method", *Science*, **306**: 1776-1780.

Karni, Edi and John A. Weymark (2024), "Impartiality and Relative Utilitarianism", *Social Choice and Welfare*, **63**: 1-18. https://doi.org/10.1007/s00355-024-01518-1

Kauppinen, Antti (2013), "Meaning and Happiness", *Philosophical Topics*, **41**: 161-185.

Krantz, David H., R. Duncan Luce, Patrick Suppes, and Amos Tversky (1971), *Foundations of Measurement: Vol. I.* Academic Press, New York.

Li, P. F. Jonah, Y. Joel Wong, and Ruth C.-L. Chao (2019), "Happiness and Meaning in Life: Unique, Differential, and Indirect Associations with Mental Health", *Counselling Psychology Quarterly*, **32**: 1-19.

Li, Chen and Peter Wakker (2024), "A Simple and General Axiomatization of Average Utility Maximization for Infinite Streams", *Journal of Economic Theory*, **216**: 1-10. https://doi.org/10.1016/j.jet.2024.105795

Lucas, Richard E., Portia S. Dyrenforth, and Ed Diener (2008), "Four Myths about Subjective Well-Being", *Social and Personality Compass*, **2**: 2001-2015.

Mancini, Anthony D., George A. Bonanno, and Andrew E. Clark (2011), "Stepping Off the Hedonic Treadmill: Individual Differences in Response to Major Life Events", *Journal of Individual Differences*, **32**: 144-152.

Marshall, Alfred (1890), *Principles of Economics.* Macmillan.

Maslow, Abraham H. (1943), "A Theory of Human Motivation", *Psychological Review*, **50**: 370-396.

Mill, John Stuart (1863), *Utilitarianism.* Parker, Son, and Bourn.

Narens, Louis (1974), "Measurement without Archimedean Axioms", Philosophy of Science, 41: 374-393.

Narveson, Jan (1973), "Moral Problems of Population", The Monist, 57: 62-86.

Nebel, Jacob M. (2022), "Totalism without Repugnance", in *Ethics and Existence: The Legacy of Derek Parfit.* Edited by Jeff McMahan, Tim Campbell, James Goodrich, and Ketan Ramakrishnan. Oxford University Press: 200-231.

Nebel, Jacob (2024), "Extensive Measurement in Social Choice", *Theoretical Economics*, **19**: 1581-1618.

Osborne, Martin J. (2025), *Models in Political Economy.* OpenBook Publishers, Cambridge, UK.

Parfit, Derek (1982), "Future Generations: Further Problems", *Philosophy and Public Affairs*, **11**: 113-172.

Parfit, Derek (1984), *Reasons and Persons.* Oxford University Press.

Parfit, Derek (1997), "Equality and Priority", Ratio, 10: 202-221.

Persson, Ingmar (2004), "The Root of the Repugnant Conclusion and its Rebuttal", in J. Ryberg and T. Tännsjö (eds.), *The Repugnant Conclusion*, Library of Ethics and Applied Philosophy: 187-200.

Pivato, Marcus (2014), "Additive Representation of Separable Preferences over Infinite Products", *Theory and Decision*, **77**: 31-83. https://doi.org/10.1007/s11238-013-9391-2

Pivato, Marcus (2020), "Rank-Additive Population Ethics", *Economic Theory*, **69**: 861-918.

Pivato, Marcus and Élise Flore Tchouante (2024), "Bayesian Social Aggregation with Non-Archimedean Utilities and Probabilities", *Economic Theory*, **77**: 561-595. https://doi.org/10.1007/s00199-023-01509-w

Rachels, Stuart (2004), "Repugnance or Intransitivity: A Repugnant but Forced Choice", in J. Ryberg and T. Tännsjö (eds.), *The Repugnant Conclusion*, Library of Ethics and Applied Philosophy: 163-186.

Rawls, John (1971), *A Theory of Justice.* Harvard University Press.

Roberts, Melinda A. (2011), "The Asymmetry: A Solution", Theoria, 77: 333-367.

Sandel, Michael J. (2012), *What Money Can't Buy: The Moral Limits of Markets.* Macmillan, New York.

Savage, Leonard J. (1954), *The Foundations of Statistics.* New York: John Wiley and Sons. (Second edition 1972, Dover.)

Schopenhauer, Arthur (1818), *The World as Will and Representation.* Translated by E. F. J. Payne. Indian Hills, Colorado: The Falcon's Wing, 1958.

Schwarz, Norbert and Gerald L. Clore (1983), "Mood, Misattribution, and Judgments of Well-Being: Informative and Directive Functions of Affective States", *Journal of Personality and Social Psychology*, **45**: 513-523.

Sikora, Richard I. (1975), "Utilitarianism: The Classical Principle and the Average Principle", Canadian Journal of Philosophy, 5: 409-419.

Smith, Adam (1759), *The Theory of Moral Sentiments.* Edited by D. D. Raphael and A. L. Macfie. Indianapolis: Liberty Fund, 1982.

Stevenson, Betsey and Justin Wolfers (2013), "Subjective Well-being and Income: Is There Any Evidence of Satiation?", *American Economic Review*, 103(3), 598–604.

Stigler, George J. and Gary S. Becker (1977), "De Gustibus Non Est Disputandum", *American Economic Review*, **67**(2): 76–90.

Strack, Fritz, Leonard L. Martin, and Norbert Schwarz (1988), "Priming and Communication: Social Determinants of Information Use in Judgments of Life Satisfaction", *European Journal of Social Psychology*, **18**: 429-442.

Strawson, Peter F. (1950), "On Referring", *Mind*, **59**: 320-344.

Tännsjö, Torbjörn (2002), "Why We Ought to Accept the Repugnant Conclusion", Utilitas, 14: 339-350.

Temkin, Larry S. (1987), "Intransitivity and the Mere Addition Paradox", Philosophy and Public Affairs, 16: 138-187.

Temkin, Larry S. (2012), Rethinking the Good: Moral Ideals and the Nature of Practical Reasoning. Oxford University Press.

Thomas, Teruji (2018), "Some Possibilities in Population Axiology", *Mind*, **127**: 807-832. https://www.jstor.org/stable/10.2307/48563945

Thornley, Elliott (2022), "A Dilemma for Lexical and Archimedean Views in Population Axiology", *Economics and Philosophy*, **38**: 395-415. https://doi.org/10.1017/S026626

United States Office of Management and Budget (2023), Circular A-94. Washington, D.C.

Veenhoven, Ruut and Floris Vergunst (2014), "The Easterlin Illusion: Economic Growth Does Go with Greater Happiness", *International Journal of Happiness and Development*, **1**: 311-343.

von Neumann, John and Oskar Morgenstern (1944), *Theory of Games and Economic Behavior*. Princeton, N.J.: Princeton University Press.

Zuber, Stéphane and Geir B. Asheim (2012), "Justifying Social Discounting: The Rank-Discounted Utilitarian Approach", Journal of Economic Theory, 147: 1572-1601.